

# PHY1163: Physics for Engineers I Module

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## Assessment Pattern

- ➡ Quizzes (10 Points)
- ➡ Assignments (10 Points)
- ➡ Practical Works (10 Points)
- ➡ Final CAT (20 Points)
- ➡ Exam (50 Points)

# Outline

## 1 INTRODUCTION

- Nature of Physical World
- Physical quantities and their Measurement
- Measurement's errors and their propagation
- Dimensions of a physical quantities

Physics is one of the most fundamental of the sciences. Scientists of all disciplines use the ideas of physics. It is also the foundation of all engineering and technology. The study of physics is an adventure. You'll find it challenging, sometimes frustrating, and often richly rewarding and satisfying.

In this section, we will see some important preliminaries that we will need throughout our study.

Having successfully completed this section, you should be able to

-  Describe the different realms of physics and their application in both natural and technological system;
-  Explain the SI unit and fundamental units that constitute it;
-  Discuss standards for measuring time, length and mass;
-  Explain the important of significant figures and handle them in calculations;
-  Discuss the measurement's errors and their propagation;
-  Apply dimensional analysis in various situations.

# Why Science?

Human history shows that people have been making continuous and serious attempts to understand what is happening around and in the World around them in general.

The inquiring mind of human tried to understand the natural phenomena by observing and analysing the environment carefully

This series of human's efforts has led us to today's modern science and technology.

# What is Science?

**Science** [From Latin *scientia*]: Knowledge.

**Science** can be defined as the knowledge gained through the systematic observations and experiments.

The **scientific methods** are *Systematic Observation, Reasoning, Modelling and Theoretical Prediction*.

A Science must have a **predictive power**, (i.e a scientific theory referring to its ability to generate testable predictions).

# What is Physics?

**Physics:** The most basic of all science.

**Physics:** The "Mother" of all sciences. Foundation of all physical sciences.

**Physics:** The Science of dealing with properties (behavior), changes of matter and energy and with the interaction between matter and energy.

**Physics** as other physical sciences (such as chemistry, astronomy and geology) is the study of non-living systems, in contrast to the [life sciences](#).

# Physics and Science

## Physics as experimental science

Physics is experimental Science based on experimental observation and quantitative measurements.

## Physics as Natural Science

The Physics studies natural phenomena in terms of Basic Laws, Principles and Quantities.

## Scope of Physics

To understand the scope of physics one needs to look at its various subdisciplines such as Classical Mechanics, Relativity, Thermodynamics, Electromagnetism, Optics and Quantum Mechanics.

# Physical Quantities

To understand natural phenomena, we must make measurements of various aspects of the nature.

The laws of Physics are expressed as mathematical relationship among physical quantities.

Physical quantities are classified into two categories: **Fundamental quantities** and **Derived quantities**.

*Fundamental quantities* are quantities which cannot be expressed in terms of any physical quantity.

*Derived quantities* are quantities that can be expressed in terms of fundamental quantities.

In Mechanics, the fundamental quantities are **Length**, **Mass** and **Time**. All other quantities in mechanics can be expressed in terms of these three.

# Standards and Units

If we are to report the result of a measurement, a **standard** must be defined.

The measurement of quantity is done by comparing it with some standard called **unit**. A unit, therefore, is any division which is accepted as one unit of that quantity.

The choice of a standard is very crucial. The chosen standard has to be *accessible, invariable, and accurate*.

In early days, many systems of units were used to measure physical quantities. To bring uniformity, the General Conference on Weights and Measures (CGPM) in the 1960s accepted the SI of unit as international system of units.

Since 1960s the SI has been periodically updated to take account of advances in science and the need for measurements in new domains.

The last major revision was adopted by the 26th CGPM (2018), which decided that the SI would be based on the fixed numerical values of a set of seven defining constants from which the definitions of the seven base units of the SI would be deduced.

The SI in the system of units in which:

- the unperturbed ground state hyperfine transition frequency of the Caesium-133 atom  $\Delta\nu_{Cs}=9\,192\,631\,770\text{ Hz}$ ,
- the speed of light in vacuum  $c$  is  $299\,792\,452\text{ m/s}$ ,
- the Planck constant  $h$  is  $6.626\,070\,634 \times 10^{-34}\text{ J s}$ ,
- the elementary charge  $e$  is  $1.602\,176\,634 \times 10^{-19}\text{ C}$ ,
- the Boltzmann constant  $k$  is  $1.380\,649 \times 10^{-23}\text{ J/K}$ ,
- the Avogadro Constant  $N_A$  is  $6.022\,140\,76 \times 10^{23}\text{ mol}^{-1}$ ,
- the luminous efficacy of monochromatic radiation of frequency  $540 \times 10^{12}\text{ Hz}$ ,  $K_{cd}$  is  $683\text{ lm/W}$

## 7 Fundamental Quantities

In the SI system, we have seven fundamental quantities.

These are:

- **Time** which the unit is the **second (s)**
- **Length** which the unit is the **meter (m)**
- **Mass** which the unit is the **kilogram (kg)**
- **Electric current** which the unit is the **ampère (A)**
- **Thermodynamic temperature** which the unit is the **kelvin (K)**
- **Amount of substance** which the unit is the **mole (mol)**
- **Luminous intensity** which the unit is the **candela (cd)**

# Decimal multiples and sub-multiples of SI units

A set of prefixes have been adopted for use with the SI units in order to express the values of quantities that are either much larger than, or much smaller than, the SI unit when used without any prefix.

| Factor    | Name  | Symbol | Factor     | Name  | Symbol |
|-----------|-------|--------|------------|-------|--------|
| $10^1$    | deca  | da     | $10^{-1}$  | deci  | d      |
| $10^2$    | hecto | h      | $10^{-2}$  | centi | c      |
| $10^3$    | kilo  | k      | $10^{-3}$  | milli | m      |
| $10^6$    | mega  | M      | $10^{-6}$  | micro | $\mu$  |
| $10^9$    | giga  | G      | $10^{-9}$  | nano  | n      |
| $10^{12}$ | tera  | T      | $10^{-12}$ | pico  | p      |
| $10^{15}$ | peta  | P      | $10^{-15}$ | femto | f      |
| $10^{18}$ | exa   | E      | $10^{-18}$ | atto  | a      |
| $10^{21}$ | zetta | Z      | $10^{-21}$ | zepto | z      |
| $10^{24}$ | yotta | Y      | $10^{-24}$ | yocto | y      |

# Rules and Conventions in measurements

For writing units of SI, here are some rules and conventions:

- The units named after scientists are not written with capital initial letter;
- The symbols of the units named after scientists should be written with capital letters;
- Small letter are used as symbols for units not derived from proper names;
- No full stop or other punctuation marks should be used within or at the end of symbols;
- The symbols of units do not take the plural form; and
- Gap space is always to be left between the number and the symbol of unit and also between the symbol for the compound units.

# Measurement errors

Experiments are never 100% accurate, always they have uncertainties in final results. The uncertainties in the measurement of physical quantities is called **error**. Error may be classified into many categories:

## Constant error

It is a constant error and it is due to fault calibration of the scale in measuring instrument.

It can be minimized by measuring by different possible methods and take the arithmetic mean value of all obtained values as the true value.

## Systematic errors

There are instrument errors and personal errors due to external sources. They occur due to certain pattern or system and they can be minimized by identifying the source of them.

# Measurement errors (Cont.)

## Gross errors

They rise due to one or more of the following reasons:

- Improper setting of the instrumentations;
- Wrong recoding of observations;
- Not taking into account the source of error and precautions;
- Using wrong values in calculations.

They can be minimized only if the observer is very careful in his observations and sincere in his approach.

## Random errors

There are very common that repeated measurements of quantity give values which are slightly different from one another. They can be minimized by repeating the measurements many times and taking the arithmetic mean of all value as a correct reading.

# Absolute and Relative uncertainties

After minimizing all there errors, we stay with the error made by the the manufacturer of the measuring instruments known as *absolute error* or *precision* of the instrument.

## Example No. 1:

Consider a simple measurement of the width of a board. Find 23.2 cm. However, measurement is only accurate to 0.1 cm (estimated). The final result will be written as  $(23.2 \pm 0.1) \text{ cm}$ .

Thus  $\pm 0.1 \text{ cm}$  is the experimental uncertainty (called margin of error or absolute error).

The percent uncertainty (called relative uncertainty or precision) is  $\pm(0.1/23.2) \times 100 \approx \pm 0.4\%$ .

# Propagation of uncertainties

Usually we use the data from measurement in mathematical operations to calculate some physical dimensions like volume, mass density, force, etc.

While using those data measurement errors are propagated in a predictable way:

## Propagation of errors in Addition and Subtraction

When adding or subtraction values the error made on the final result is equal to the sum of individual errors,

$$x = A \pm B, \quad (1)$$

the error on  $x$  is given by

$$\Delta x = \Delta A + \Delta B \quad (2)$$

# Propagation of uncertainties (Cont.)

## Propagation of errors in Multiplication and Division

When multiplying or dividing values, the error propagation is calculated following this rule:

$$x = A \times B \text{ or } x = \frac{A}{B}, \quad (3)$$

the error on  $x$  is calculated by

$$\frac{\Delta x}{x} = \frac{\Delta A}{A} + \frac{\Delta B}{B} \quad (4)$$

# Uncertainties (Examples)

## Example No. 2

In an experiment for determination of the density  $\rho$  of a rectangular block of a metal, the dimension of the block are measured with a callipers having a least count of 0.01 cm and its mass is measured with beam balance of least count 0.1 g.

The measured values are:

Mass of the block ( $m$ )=39.3 g,

Length of the block ( $x$ )=5.12 cm,

Breadth of the block ( $y$ )=2.56 cm,

Thickness of the block ( $z$ )=0.37 cm.

What is the value of the density found after calculations?

# Significant figures

Significant figure ("sig figs") also called significant digits is the number of reliably known digits in a number. ie number of meaningful figures in the measured value.

## Non-zero digits

All non-zero digits are significant. Example: 1.12 has 3 significant digits(1, 1 and 2), 9456 has 4 significant digits(9, 4, 5 and 6).

## Leading zeros

All zeros to the left of the first non-zero digit are not significant. Example: in 0.005 only 5 is significant, the 0.0000506 has 3 significant digits (5,0,6).

## Significant figures (Cont.)

### Tailing zeros

Zeros to the right of the last non-zero digit are significant if the decimal point is shown. Example: 4.00 has 3 significant digits, 300.000 has 6 significant digits, 0.0560 has 3 significant digits and 0.0050600 has 5 significant digits (5,0,6,0,0).

Zeros to the right of the non-zero digit (tailing zeros) are not significant if there is an understood decimal point. Example: in 600 only, 6 is significant digits; in 9800 000, only 9 and 8 are significant.

### Scientific notation

In scientific notation, the exponent part of the number is not significant. Example:  $3.45 \times 10^3$  has only 3 significant digits (3, 4 and 5),  $0.05600 \times 10^6$  has only 4 significant digits (5, 6, 0 and 0)

## Significant figures (Cont.)

### Significant figures in multiplication (division) operations

When approximate numbers are multiplied or divided, the number of significant digits in the final answer is the same as the number of significant digits in the least accurate of the factors.

### Significant figures in addition (subtraction) operations

When approximate numbers are added or subtracted, the number of decimal places should equal the smallest number of decimal places of any term in the sum or difference.

# Rounding numbers

The result of calculation with the number containing more than one uncertain digit should be rounded off.

## Rule 1

When rounding, if the remainder beyond the last digit to be reported is less than 5 drop the last digit.

## Rule 2

If the remainder is greater than 5, increase the final digit by 1.

## Rule 3

To prevent rounding bias, if the remainder is exactly 5, then round the last digit to the closest even number.

# Rounding numbers(Examples)

## Example No. 3

Round the following to 3 significant figures:

- 4.99499
- 0.09403
- 95,632
- 0.02032
- 2.3452
- 0.08757
- 23,650.01
- 4.99502
- 3.77500
- 0.024450
- 96,650
- 5.09500

# Dimensional Analysis

The dimensions of a physical quantity are the power to which the fundamental quantities (length, time, mass) must be raised to form its units.

Square brackets [ ] are usually used to denote the dimensions of a quantity. The three fundamental quantities of length, time and mass have dimension [L], [T] and [M] respectively. Note that it is the capital letters that are used to denote the dimensions.

The derived units of all the physical quantities can be obtained from the seven fundamental units of length, mass, time, intensity of current, temperature, amount of substance and luminosity intensity. Dimensions are helpful in working out and verifying relationships.

# Dimensional Analysis (Cont.)

An equation is dimensionally correct if the dimensions of the various terms on either side of the equation are the same.

This is based on the fact that only two quantities of the same dimension can be added up, the resulting quantity also possessing the same dimensions.

Note that the correctness of the dimensions of an equation does not mean the correctness of physical quantity.

# Dimensional Analysis (Example)

## Example No. 4

The speed of sound in a gas might plausible depend on the pressure  $p$ , the density  $\rho$  and the volume  $V$  of the gas.

Use dimensional analysis to determine the exponents  $x$ ,  $y$ , *and*  $z$  in the formula:

$$v = Cp^x \rho^y V^z \quad (5)$$

Where  $C$  is a dimensionless constant. Incidentally, the MKS units of pressure are kilograms per meter per second squared.

# Outline

## 2 VECTOR ALGEBRA

- Vectors and Scalars
- Addition and Subtraction of Vectors
- Vector Multiplication

In Physics, we often need to work with physical quantities that have numerical and directional properties. Quantities of this nature are vectors. This section is primarily concerned with general properties of vector quantities. We discuss the addition and subtraction of vectors, together with some common applications to physical situations.

Vectors are used throughout this module. Therefore, it is imperative that you master the techniques discussed in this section.

Having successfully completed this section, you should be able to

-  Distinguish scalar quantities from vector quantities;
-  Represent vectors graphically and give their properties;
-  Add and subtract vectors and multiply them by scalars;
-  Describe the components of vector and use them in calculations;
-  Operate the two ways of multiplying vectors.

# Scalar and Vector quantities

## Scalars

**A scalar** is a physical quantity that has only magnitude (size) and can be completely defined by a number and a unit.

Examples of scalars are time, mass, volume, temperature, pressure, density, electrical charge, work, frequency,...

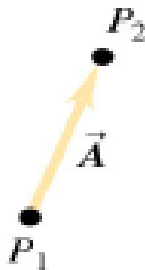
## Vectors

**A vector** is a physical quantity that has both magnitude and direction.

Examples of vectors are displacement, velocity, force,...

# Representation of a vector

Vectors are represented pictorially by an arrow from one point to another, and they are represented symbolically by a letter with an arrow above it.



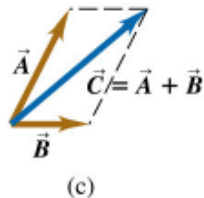
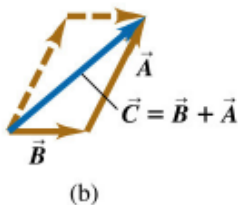
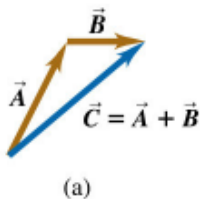
## Some vector properties

Vectors observe some properties which are:

- Two vectors that have the same direction are said to be parallel.
- Two vectors that have opposite direction are said to be anti-parallel.
- Two vectors that have the same length and the same direction are said to be equal no matter where they are located.
- The negative of a vector is a vector with the same magnitude but opposite direction.

# Addition of vectors

Given two vectors  $\vec{A}$  and  $\vec{B}$ , their sum (or their resultant) is the vector  $\vec{C}$  obtained by joining the tail of one of them to the head of another when the head of the first is coinciding with the tail of the second vector.

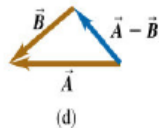
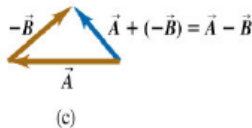
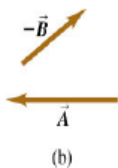
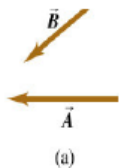


Vector addition is commutative (the order of a vector addition doesn't matter).

# Subtraction of Vectors

The operation of vector subtraction makes use of the definition of the negative of vector;

$$\vec{A} - \vec{B} = \vec{A} + (-\vec{B})$$

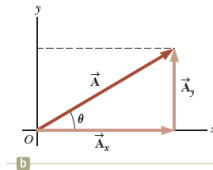
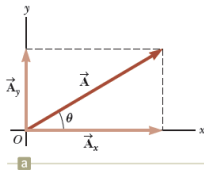


# Components of a vector

## Rectangular components

One of the method of adding vectors makes use of the projections of vectors along coordinate axes.

These projections are called the components of the vector or its rectangular components. Any vector can be completely described by its components.



## Components of a vector (Cont.)

### Components of vectors

The component  $A_x$  represent the projection of  $\vec{A}$  along  $x$  axis, and the component  $A_y$  represent the projection of  $\vec{A}$  along  $y$  axis.

### Components and trigonometric relations

The definition of sine and cosine, we see that  $\cos \theta = \frac{A_x}{A}$  and that  $\sin \theta = \frac{A_y}{A}$ . Hence, the component of  $\vec{A}$  are

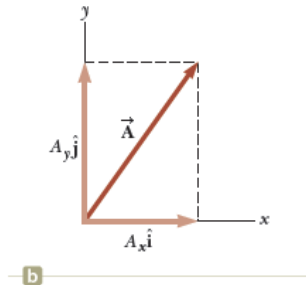
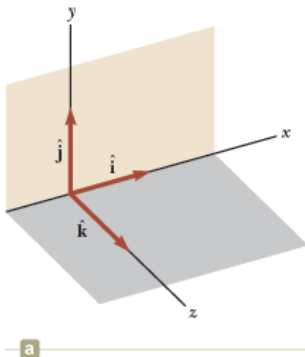
$$A_x = A \cos \theta \quad (6)$$

$$A_y = A \sin \theta \quad (7)$$

# Unit vector

Vector quantities often are expressed in terms of unit vectors.

A unit vector is a dimensionless vector having a magnitude of exactly 1.



## A vector in unit vector notation

The product of the components  $A_x$  and the unity vector  $\hat{i}$  is the component vector  $\vec{A}_x = A_x \hat{i}$ , which lies on the  $x$  axis and has magnitude  $|A_x|$ .

Likewise,  $\vec{A}_y = A_y \hat{j}$  is the component vector of magnitude  $|A_y|$  lying on the  $y$  axis.

The unit-vector notation for the vector  $\vec{A}$  is

$$\vec{A} = A_x \hat{i} + A_y \hat{j} \quad (8)$$

The sum of vectors  $\vec{A}$  and  $\vec{B}$  is given by  $\vec{R} = (A_x + B_x)\hat{i} + (A_y + B_y)\hat{j}$ . Because  $\vec{R} = R_x \hat{i} + R_y \hat{j}$ , we see that the components of the resultant vector are  $R_x = A_x + B_x$  and  $R_y = A_y + B_y$ .

## Vectors (Examples)

### Example No. 5

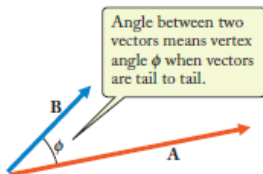
A hiker begins a trip by first walking 25.0 km southeast from her car. She stops and sets up her tent for night. On the second day, she walks 40.0 km in a direction  $60.0^\circ$  north of east, at which point she discovers a forest ranger's tower.

- (A) Determine the components of the hiker's displacement for each day.
- (B) Determine the components of the hiker's resultant displacement for the trip. Find an expression for the displacement in terms of unit vectors.
- (C) After reaching the tower, the hiker wishes to return to her car along a single straight line. What are the components of the vector representing this hike? What should the direction of the hike be?

# Dot product

The dot product (also called the scalar product) of two vectors  $\vec{A}$  and  $\vec{B}$  is denoted by  $\vec{A} \cdot \vec{B}$

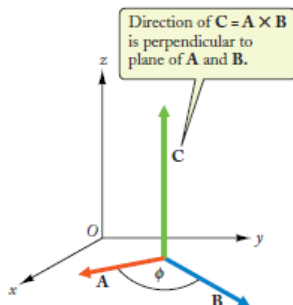
This quantity is simply *the product of the magnitudes of the two vectors and the cosine of the angle between them.*



$$\vec{A} \cdot \vec{B} = (A_x B_x) + (A_y B_y) = |\vec{A}| |\vec{B}| \cos \phi \quad (9)$$

# Cross product

In contrast to the dot product of two vectors, which is a scalar, the cross product (also called vector product) of two vectors is a vector.



## Cross product (Cont.)

We write the vector resulting from the cross product as

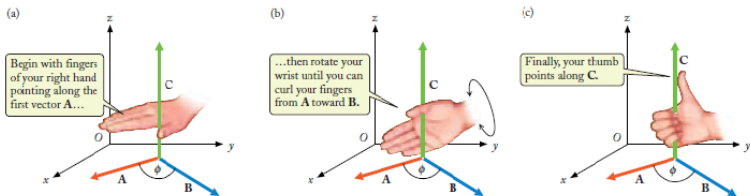
$$\vec{C} = \vec{A} \times \vec{B} \quad (10)$$

then the magnitude of this vector is

$$|\vec{C}| = |\vec{A}||\vec{B}| \sin \phi \quad (11)$$

# Right hand rule for cross product

The direction of  $\vec{C}$  along this perpendicular is given by the right-hand rule: put the fingers of your right hand along  $\vec{A}$ , and curl them toward  $\vec{B}$  in the direction of the small angle from  $\vec{A}$  to  $\vec{B}$ , the thumb then points along  $\vec{C}$ .



Note that the finger must be curled from the first vector in the product toward the second.

## Cross product of unit perpendicular vectors

The cross product of two unit vectors is particular simple: since each vector has unit magnitude, application of the previous theories gives

$$\hat{i} \times \hat{i} = 0; \hat{i} \times \hat{j} = \hat{k}; \text{ and, } \hat{i} \times \hat{k} = -\hat{j}$$

Then,

$$\hat{j} \times \hat{i} = -\hat{k}; \hat{j} \times \hat{j} = 0 \text{ and, } \hat{j} \times \hat{k} = \hat{i}$$

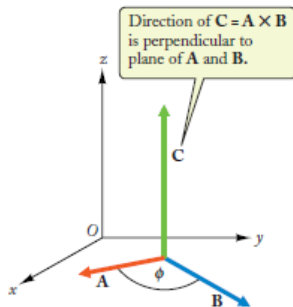
and

$$\hat{k} \times \hat{i} = \hat{j}; \hat{k} \times \hat{j} = -\hat{i}; \text{ and, } \hat{k} \times \hat{k} = 0$$

# Cross product of any two vectors

Now using this technique on the cross product vectors  $\vec{A} \times \vec{B}$ , with  $\vec{A} = A_x\hat{i} + A_y\hat{j} + A_z\hat{k}$  and  $\vec{B} = B_x\hat{i} + B_y\hat{j} + B_z\hat{k}$  gives

$$\vec{A} \times \vec{B} = \vec{C} = (A_y B_z - A_z B_y)\hat{i} + (A_z B_x - A_x B_z)\hat{j} + (A_x B_y - A_y B_x)\hat{k} \quad (12)$$



# Vector Multiplication (Examples)

## Example No. 6

- 1 Two sides of a triangle are formed by vectors  $\vec{A} = 3\vec{i} + 6\vec{j} - 2\vec{k}$  and  $\vec{B} = 4\vec{i} - \vec{j} + 3\vec{k}$ . Determine all angles of the triangle.
- 2 The diagonals of a parallelogram are given by vectors  $(3\vec{i} + \vec{j} + 2\vec{k}) m$  and  $(\vec{i} - 3\vec{j} + 4\vec{k}) m$ . Find the area of the parallelogram.
- 3 A farmland, located on a hill, has triangular shape of corners A(1,1,0), B(0,2,4) and C(-1,-1,1). (Values given in hundreds of meters). Find the area of the farmland and express the result in hectares.

# Triple Product

Consider vectors described in a rectangular coordinate system as  $\vec{a} = a_x i + a_y j + a_z k$  ;  $\vec{b} = b_x i + b_y j + b_z k$  and  $\vec{c} = c_x i + c_y j + c_z k$ .

The triple product can be evaluated by using the relation:

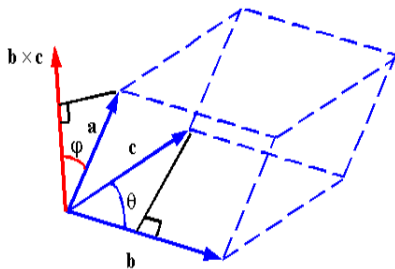
$$\vec{a} \bullet (\vec{b} \times \vec{c}) = \alpha = \begin{vmatrix} a_x & a_y & a_z \\ b_x & b_y & b_z \\ c_x & c_y & c_z \end{vmatrix} \quad (13)$$

The triple product has the following properties:

$$\vec{a} \bullet (\vec{b} \times \vec{c}) = (\vec{b} \times \vec{c}) \bullet \vec{a} = \vec{c} \bullet (\vec{a} \times \vec{b}) = \vec{b} \bullet (\vec{c} \times \vec{a})$$

## Triple Product (Cont.)

The value of the triple product of vectors **a**, **b**, and **c** is equal to the *volume* of the parallelepiped constructed from the vectors. This can be seen from the figure since  $Volume = abc \sin\theta \cos\varphi$ .



$$\vec{a} \bullet (\vec{b} \times \vec{c}) = a|\vec{b} \times \vec{c}| \cos\varphi = abc \sin\theta \cos\varphi \quad (14)$$

# Outline

## 3 KINEMATICS

- Introduction
- Motion in one dimension
- Motion in Two Dimensions
- Uniform Circular Motion
- Rotational Motion

Electrons swarming around atomic nuclei, cars speeding along a highway, blood coursing through your veins, galaxies rushing apart in the expanding universe— all these are examples of matter in motion. The study of motion without regard to its causes is called **kinematics**. This section deals with kinematics. We will start with a single object moving in straight line, then to more complex and more interesting situations of the motion in two dimensions.

Having successfully completed this section, you should be able to:

-  Explain the meaning of and relationships among displacement, velocity, and acceleration in one- and two-dimensional motion;
-  Calculate average and instantaneous velocities and accelerations;
-  Interpret graphs of position versus time, velocity versus time, and acceleration versus time for one-dimensional motion;
-  Transform velocities to different reference frames;
-  Solving quantitative problems involving motion in one and two dimensions with constant acceleration.

# Frame of Reference

## Concept of frame of reference

A reference frame is any system relative to which an event in space and time can be described.

It is a coordinate system with a clock.

We can identify two types of reference frames:

- Inertial reference frame; and
- Non-inertial reference frame.

Inertial reference frames are reference frames which are not accelerating or rotating, while Non-inertial reference frames are accelerating or rotating.

## Kinematics

Kinematics is the part of Mechanics which deals with the motion of a body without reference to force and mass.

# Displacement

When an object moves along a straight line, we describe its position with respect to an origin  $O$  by means of a coordinate such as  $x$ .

If the object starts at position  $x_1$  at time  $t_1$  and arrives at position  $x_2$  at time  $t_2$ , the object's **displacement** is a vector quantity whose  $x$  component is  $\Delta x = x_2 - x_1$



The displacement doesn't depend on the details of how the object travels between  $x_1$  and  $x_2$

## Average and Instantaneous Velocity

The rate of change of position with respect to time is given by the *average velocity*, a vector quantity whose  $x$  component is

$$v_{av,x} = \frac{x_2 - x_1}{t_2 - t_1} = \frac{\Delta x}{\Delta t} \quad (15)$$

As the average velocity is calculated for progressively smaller intervals of time  $\Delta t$ , the average velocity approaches the *instantaneous velocity*  $v_x$ :

$$v_x = \lim_{\Delta t \rightarrow 0} \frac{\Delta x}{\Delta t} = \frac{dx}{dt} \quad (16)$$

## Average and Instantaneous Acceleration

When the velocity of an object changes with time, we say that the object has an *acceleration*.

For an object with velocity  $v_{1x}$  at time  $t_1$  and velocity  $v_{2x}$  at time  $t_2$ , the  $x$  component of *average acceleration* is

$$a_{av,x} = \frac{v_{2x} - v_{1x}}{t_2 - t_1} = \frac{\Delta v}{\Delta t} \quad (17)$$

As with velocity, when the average acceleration is calculated for smaller intervals of time  $\Delta t$ , the average acceleration approaches the *instantaneous acceleration*  $a_x$ .

$$a_x = \lim_{\Delta t \rightarrow 0} \frac{\Delta v}{\Delta t} = \frac{dv}{dt} \quad (18)$$

# Equations of Motion

When an object moves with constant acceleration in a straight line, the following equations describe its position and velocity as functions of time:

$$x = x_0 + v_{0x}t + \frac{1}{2}a_x t^2 \quad (19)$$

and

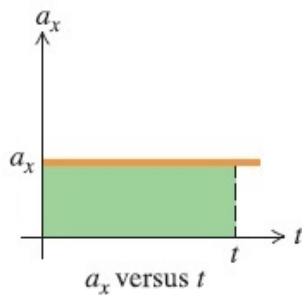
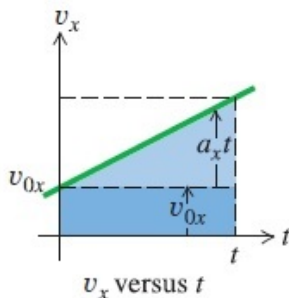
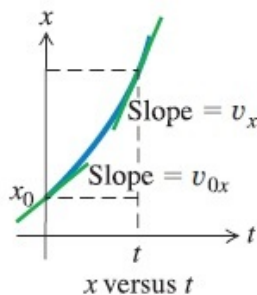
$$v_x = v_{0x} + a_x t \quad (20)$$

In these equations,  $x_0$  and  $v_{0x}$  are, respectively, the position and the velocity at an initial time  $t = 0$ , and  $x$  and  $v_x$  are, respectively, the position and the velocity at any later time  $t$ .

The following equation relates velocity, acceleration, and position without explicit reference to time:

$$v_x^2 = v_{0x}^2 + 2a_x(x - x_0) \quad (21)$$

# Graphic representation of displacement, velocity and acceleration versus time



# Free-Fall Motion

A freely falling object is an object that moves under the influence of a constant gravitational force.

The term *free fall* includes objects that are initially at rest, as well as objects that have an initial upward or downward velocity.

When the effects of air resistance are excluded, all bodies at a particular location fall with the same acceleration, regardless of their size or weight.

The constant acceleration of freely falling objects is called the acceleration due to gravity, and we denote its magnitude with the letter  $g$ .

At or near the earth's surface, the value of  $g$  is approximately  $9.81\text{ms}^{-2}$

The free fall motion is treated as an accelerated (or decelerated) motion.

# 1D Motion (Examples)

## Example No. 7

- ① Compute your average velocity in the following two cases: (a) You walk 73.2 m at a speed of 1.22 m/s and then run 13.2 m at a speed of 3.02 m/s along a straight track. (b) You walk for 1.00 min at a speed of 1.22 m/s and then run for 1.00 min at 3.05 m/s along a straight track. (c) Graph  $x$  versus  $t$  for both cases and indicate how the average velocity is found on the graph.
- ② A car initially traveling a straight stretch of a highway at 15 m/s accelerate with a constant acceleration of  $2.0 \text{ m/s}^2$  in order to pass a truck. (a) What is the velocity of the car after 5.0 s? (b) What distance does the car travel during its 5.0 s of acceleration?
- ③ (a) With what speed must a ball be thrown vertically from ground level to rise to a maximum height of 50 m? (b) How long will be in the air? (c) Sketch graphs of  $y$ ,  $v$  and  $a$  versus  $t$  for the ball. On the first two graph, indicate the time at which 50 m is reached.

# Vector Position

## Position vector

The position vector  $\vec{r}$  of an object in a plane is the displacement vector from the origin to that object.

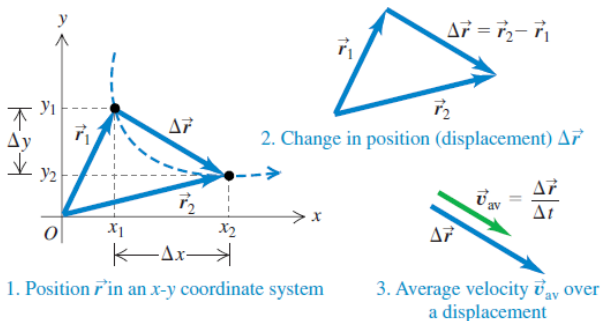
Its components are the coordinates  $x$  and  $y$ .

The average velocity  $\vec{v}_{av}$  of the object during a time interval  $\Delta t$  is its displacement  $\Delta \vec{r}$  divided by  $\Delta t$ .

## Average velocity

$$\vec{v}_{av} = \frac{\vec{r}_2 - \vec{r}_1}{t_2 - t_1} = \frac{\Delta \vec{r}}{\Delta t} \quad (22)$$

# Graphic representations



Because  $\Delta t$  is a scalar quantity, the direction of average velocity vector  $\vec{v}_{av}$  is determined entirely by the direction of the vector displacement  $\Delta \vec{r}$ .

# Instantaneous Velocity

The instantaneous velocity  $\vec{v}$  is given by

$$\vec{v} = \lim_{\Delta t \rightarrow 0} \frac{\Delta \vec{r}}{\Delta t} = \frac{d\vec{r}}{dt} \quad (23)$$

its components:

$$v_x = \lim_{\Delta t \rightarrow 0} \frac{\Delta x}{\Delta t} = \frac{dx}{dt} \quad (24a)$$

$$v_y = \lim_{\Delta t \rightarrow 0} \frac{\Delta y}{\Delta t} = \frac{dy}{dt} \quad (24b)$$

# Acceleration

The average acceleration,  $\vec{a}_{av}$ , during the time interval  $\Delta t$  is the change in velocity,  $\Delta \vec{v}$ , divided by the time interval  $\Delta t$ :

$$\vec{a}_{av} = \frac{\vec{v}_2 - \vec{v}_1}{t_1 - t_2} = \frac{\Delta \vec{v}}{\Delta t} \quad (25)$$

The instantaneous acceleration is given by

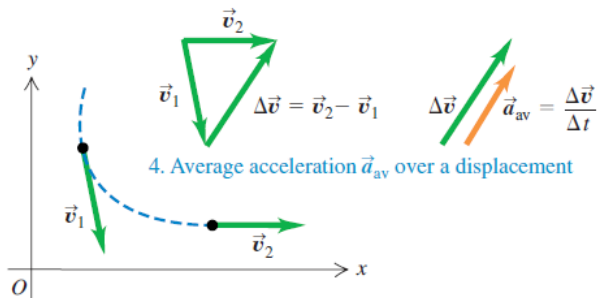
$$\vec{a} = \lim_{\Delta t \rightarrow 0} \frac{\Delta \vec{v}}{\Delta t} = \frac{d\vec{v}}{dt} \quad (26)$$

and its components:

$$a_x = \lim_{\Delta t \rightarrow 0} \frac{\Delta v_x}{\Delta t} \quad (27a)$$

$$a_y = \lim_{\Delta t \rightarrow 0} \frac{\Delta v_y}{\Delta t} \quad (27b)$$

# Graphic representation of velocity and acceleration

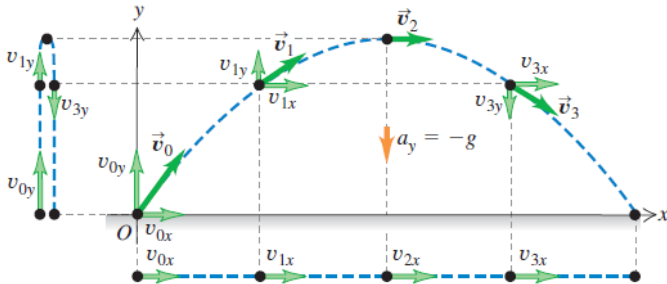


An object has acceleration if *either* its speed or its direction of motion changes—that is, if *either* the magnitude or direction of its velocity changes.

# Projectile Motion

Projectile motion occurs when an object is given an initial velocity and then follows a path determined entirely by the effect of a constant gravitational force.

The path, called a trajectory, is a parabola in the  $x - y$  plane. The vertical motion of a projectile is independent of its horizontal motion.



The vertical and horizontal components of a projectile's motion are independent.

## Projectile Motion (Cont.)

In projectile motion  $a_x = 0$  (there is no horizontal component of acceleration) and  $a_y = -g$  (a constant vertical component of acceleration due to a constant gravitational force).

The coordinates components, as functions of time, are

$$x = (v_o \cos \theta_o)t, \quad (28a)$$

$$y = (v_o \sin \theta_o)t - \frac{1}{2}gt^2. \quad (28b)$$

and velocity components, are

$$v_x = (v_o \cos \theta_o), \quad (29a)$$

$$v_y = (v_o \sin \theta_o) - gt. \quad (29b)$$

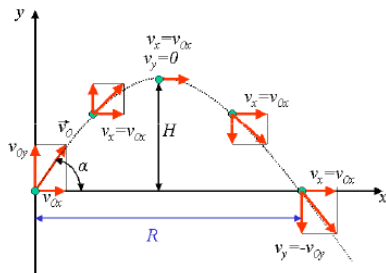
# Trajectory, Maximum Height and Horizontal Range

The trajectory, the maximum height and the horizontal range, reached by an object under projectile motion, are respectively given by

$$y = x \tan \theta - \frac{g}{2v_0^2 \cos^2 \theta} x^2, \quad (30a)$$

$$H = \frac{v_o^2 \sin^2 \theta}{2g}, \quad (30b)$$

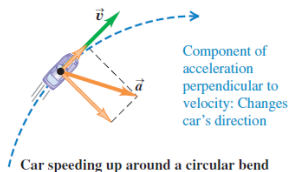
$$R = \frac{v_o^2 \sin 2\theta}{g}. \quad (30c)$$



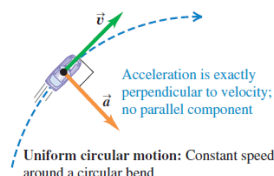
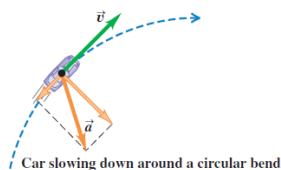
# Introduction

When a particle moves in circle with constant speed, the motion is called *uniform circular motion*.

Component of acceleration parallel to velocity:  
Changes car's speed



Component of acceleration  
perpendicular to velocity: Changes  
car's direction

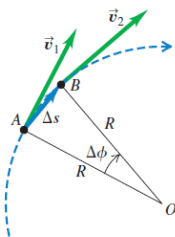


The component of acceleration perpendicular to the path, which causes the direction of the velocity to change, is related in a simple way to the speed  $v$  of the particle and the radius  $R$  of the circle.

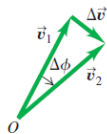
# Acceleration in a uniform circular motion

The magnitude  $a_{av}$  of the average acceleration  $\vec{a}_{av}$  during  $\Delta t$

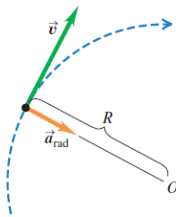
$$a_{av} = \frac{|\Delta \vec{v}|}{\Delta t} = \frac{v_1}{R} \frac{\Delta s}{\Delta t} \quad (31)$$



(a) A point moves a distance  $\Delta s$  at constant speed along a circular path.



(b) The corresponding change in velocity.



(c) The acceleration in uniform circular motion always points toward the center of the circle.

## Acceleration in a uniform circular motion (Cont.)

The magnitude  $a$  of the instantaneous acceleration  $\vec{a}$  at point  $A$  is the limit of this expression as  $\Delta t$  approaches zero and  $B$  gets closer and closer to point  $A$ .

$$a = \frac{v_1}{R} \frac{\Delta s}{\Delta t} = \frac{v_1}{R} \lim_{\Delta t \rightarrow 0} \frac{\Delta s}{\Delta t} \quad (32)$$

Thus, *"When a particle moves in a circular path with radius  $R$  and with constant speed  $v$ , it has an acceleration with magnitude*

$$a = \frac{v^2}{R} \quad (33)$$

*always directed toward the center of the circle and perpendicular to the instantaneous velocity  $v$  at each instant".*

# Angular Velocity

## Introduction

Just as we described straight-line motion in terms of the rate of change of the coordinate  $x$ , we can describe rotational motion of a body in terms of the rate of the rate of change of  $\theta$ .

## Average Angular Velocity

When a rigid body rotates through an angular displacement  $\Delta\theta = \theta_2 - \theta_1$  in a time interval  $\Delta t = t_2 - t_1$ , the average angular velocity  $\omega_{av}$  is defined as the ratio of angular displacement to the time interval:

$$\omega_{av} = \frac{\theta_2 - \theta_1}{t_2 - t_1} = \frac{\Delta\theta}{\Delta t} \quad (34)$$

## Angular Velocity (Cont.)

We can then define the *instantaneous angular velocity*  $\omega$  as the limit of  $\omega_{av}$  as  $\Delta t$  approaches zero:

$$\omega = \lim_{\Delta t \rightarrow 0} \frac{\Delta \theta}{\Delta t} \quad (35)$$

# Angular Acceleration

When the angular velocity of the body changes, the body has an angular acceleration.

We define average and instantaneous angular accelerations as follows

Average angular acceleration

$$\alpha_{av} = \frac{\omega_2 - \omega_1}{t_2 - t_1} = \frac{\Delta\omega}{\Delta t} \quad (36)$$

Instantaneous angular acceleration

$$\alpha = \lim_{\Delta t \rightarrow 0} \frac{\Delta\omega}{\Delta t} \quad (37)$$

## Equations of motion

When a body with constant angular acceleration, the angular position, velocity and acceleration are related by the equations:

$$\omega = \omega_0 + \alpha t \quad (38a)$$

$$\theta = \theta_0 + \omega_0 t + \frac{1}{2} \alpha t^2 \quad (38b)$$

$$\omega^2 = \omega_0^2 + 2\alpha(\theta - \theta_0) \quad (38c)$$

## 2D Motion (Examples)

### Example No. 8

- ❶ A ski jumper leaves the ski track moving in the horizontal direction with a speed of  $25.0 \text{ m/s}$ . The landing incline below her falls off with a slope of  $35.0^\circ$ . Where does she land on the incline?
- ❷ A projectile is in the air for  $10 \text{ s}$  and strikes the ground  $255 \text{ m}$  from its starting point. (a) Calculate the components of its initial velocity. (b) Determine the magnitude and direction of the velocity at the highest point of its trajectory. (c)  $8 \text{ s}$  after firing. Determine: (i) The position of the projectile, (ii) The components of the velocity of the projectile, (iii) The magnitude and direction of the acceleration of the projectile.
- ❸ A tire placed on a balancing machine in a service station starts from rest and turns through  $5.3$  revolution in  $2.3 \text{ s}$  before reaching its final angular speed. What is the angular acceleration of the tire (Assuming that this quantity remains constant)? What is the final speed of the tire?

# Outline

## 4 DYNAMICS

- Introduction
- Newton's Laws of Motion
- Application of Newton's laws of motion
- Uniform Circular Motion and Gravitation
- Work, Energy and Power
- Linear Momentum
- Collisions
- Impulse
- Center of the Mass

In the previous section, we have discussed how to use the language and mathematics of mathematics to describe motion in one, two or three dimensions. But what causes bodies to move the way that they do? What might cause one body to remain at rest and another to accelerate? The answers to these and similar questions take us into the subject of **dynamics**. This section discusses the principles of dynamics. We deal with the Newton's Laws of motion and their application, Uniform and circular motion and gravitation, Work–Energy and Power and we discuss the concept of linear momentum and collision.

Having successfully completed this section, you should be able to:

-  Explain the concept of force and its role in causing change in motion;
-  Describe the fundamental forces of the physics;
-  State Newton's laws of motion and apply them to one– and two–dimensional motion;
-  Explain the concept of work and evaluate the work done different forces;
-  Describe the concepts of energy and the relation between energy and power.

# Introduction

Dynamics is the part of classical mechanics concerned with forces that produce changes in motion or changes in other properties, such as the shape and the size of objects.

This leads us to the concepts of the force and the mass and the laws that govern the motion of an object.

Thus, Dynamics study the motion of objects and the causes and effects of that motion.

# Useful concepts

## Mass

*The mass* of an object is a measure of the inertia of the object.

For several centuries, Physicist have found it useful to think of mass as representation of the amount of or quantity-of-matter.

## Inertia

*Inertia* is the tendency of a body at rest to remain at rest, and of the body in motion to continue moving with unchanged velocity.

## Force

*Force*, a vector quantity, is a quantitative measure of the mechanical interaction between two objects.

# Concept of force

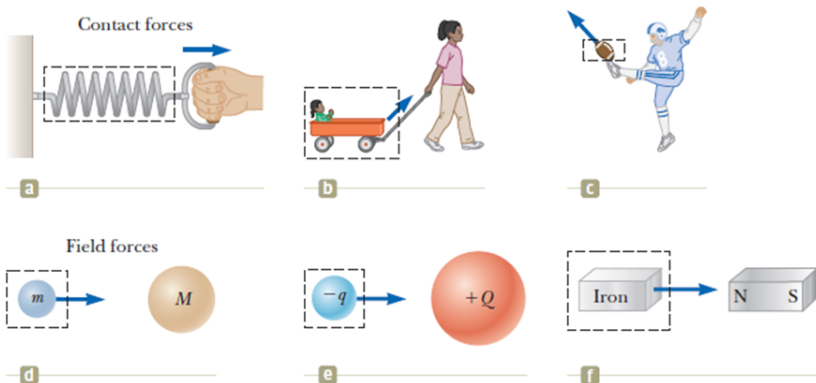
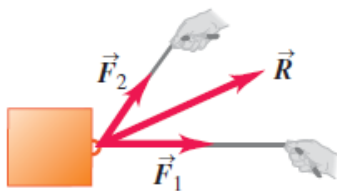
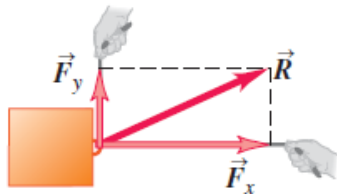


Figure 1: *Field force and contact force.*

# Graphic representation of a force



Two forces acting on a point have the same effect as a single force  $\vec{R}$  equal to their vector sum.



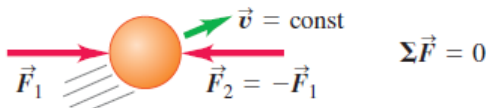
A force may be treated as the vector sum of its component vectors.

Figure 2: Graph representation of forces acting on an object.

# Newton's First law of Motion

## Statement of the First Law of Motion

When no force acts on an object or when the vector sum of all forces acting on it is zero, the object is in equilibrium. If the object is initially at rest, it remains at rest; if it is initially in motion, it continues to move with constant velocity.



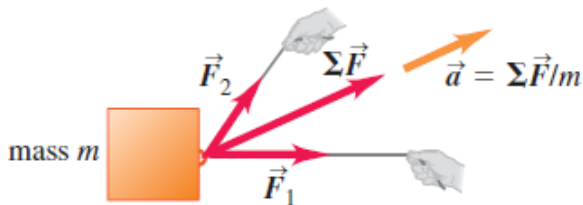
If the vector sum of the forces acting on an object is zero, the object remains at rest or (as here) in motion with constant velocity.

# Newton's Second Law of Motion

As we said, the inertial properties of an object are characterized by its mass.

## Statement of the Second Law of Motion

The acceleration of an object under the action of a given set of forces is directly proportional to the vector sum of the forces and inversely proportional to the mass of the object.

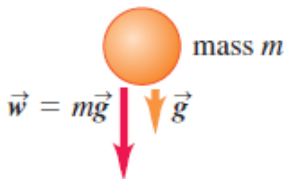


An object's acceleration depends on its mass and on the net force acting on it.

# Mass and Weight

## Differentiate the Mass from the Weight of an object

The weight of an object is the gravitational force exerted on it by the earth (or other object exerts the gravitational force). Weight is a force. The magnitude of the weight of an object at any specific location is equal to the product of the mass of the object and the magnitude of the acceleration due to gravity at that location.



An object's weight depends on its mass and on the acceleration due to gravity at its location.

# Newton's Third Law of Motion

## Statement of the Third Law of Motion

When two bodies interact, they exert forces on each other that, at each instant, are equal in magnitude and opposite in direction. The two forces in an action-reaction pair always act on two different bodies; they never act on the same object.



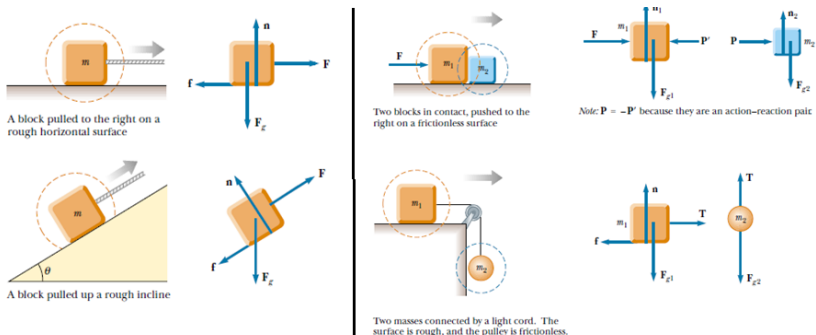
An action–reaction pair:  $\vec{F}_{\text{ball on foot}} = -\vec{F}_{\text{foot on ball}}$ .  
The two forces represent a *mutual interaction* of two objects, and each acts on a *different* object.

# Free-Body Diagrams

## Illustration

A free-body diagram is a diagram showing the chosen object by itself, "free" of its surroundings, with vectors drawn to show the forces applied to it by the various other objects that interact with it.

A free-body diagram illustrates all the external forces acting on an object.



# Application of the First Law

Newton's three laws of motion, the foundation of classical mechanics, can be stated very simply, as we have seen. But applying these laws requires some analytical skills and some problem-solving technique.

## Equilibrium of a Particle

When an object is in equilibrium, the vector sum of the forces acting on it must be zero  $\sum \vec{F} = 0$ . In component form,

$$\sum F_x = 0, \quad \sum F_y = 0 \quad (39)$$

Free-body diagrams are useful in identifying the forces acting on the object being considered. Newton's third law is also frequently needed in equilibrium problems.

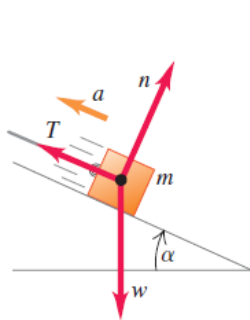
**Remember!** The two forces in an action–reaction pair never act on the same object.

# Application of the Second Law

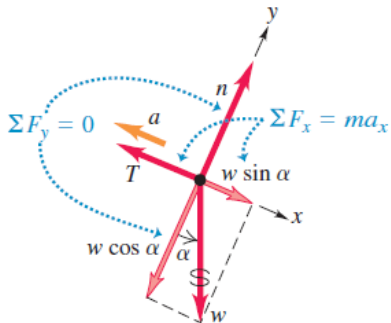
When the vector sum of forces on an object is not zero, the object has an acceleration determined by Newton's second law,  $\sum \vec{F} = m\vec{a}$ .

In component form,

$$\sum F_x = ma_x, \quad \sum F_y = ma_y \quad (40)$$



$$\sum \vec{F} = m\vec{a}.$$



A free-body diagram for the object.

## Contact Forces and Friction

The contact force between two objects can always be represented in terms of a normal component  $n$  perpendicular to the surface of interaction and a frictional component  $f$  parallel to the surface.

When sliding occurs, the kinetic-friction force  $f_k$  is often approximately proportional to the normal force,  $n$ .

Then the proportionality constant is  $\mu_k$ , the coefficient of kinetic friction:

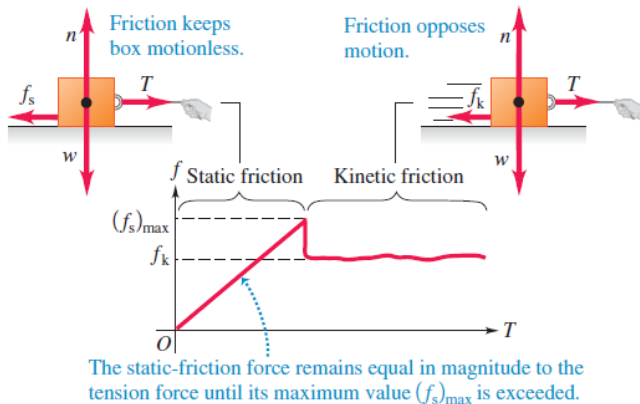
$$f_k = \mu_k n \quad (41)$$

When there is no relative motion, the maximum possible friction force is approximately proportional to the normal force, and the proportionality constant is  $\mu_s$ , the coefficient of static friction. The governing equation is

$$f_k \leq \mu_s n \quad (42)$$

# Contact Forces and Friction (Cont.)

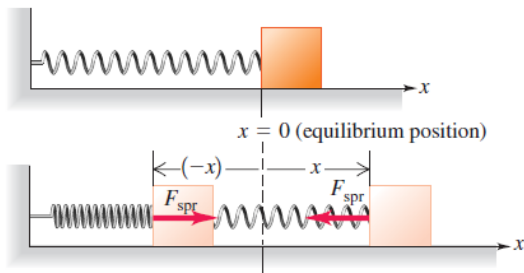
The actual static-friction force may be anything from zero to the maximum value given by the Equation 42, depend on the situation. Usually  $\mu_k$  is less than  $\mu_s$  for a given pair of surfaces.



# Elastic Forces

When forces act on a solid object, the object usually deforms. In some cases, such as a stretched or compressed spring, the deformation is approximately proportional to the magnitude of the applied force. This proportionality is called Hooke's law:

$$F_{spr} = -kx \quad (43)$$



When stretched or compressed from equilibrium, the spring exerts a force  $\vec{F}_{spr}$  whose magnitude is approximately proportional to the displacement.

# Dynamics of Circular Motion

## Introduction

Objects move in circular paths in a wide variety of situations, such as cars on racetracks, rotating machine parts, and the motions of planets and satellites, so it's worthwhile to study this special class of motions in detail.

For an object in uniform circular motion, the acceleration vector  $\vec{a}_{rad}$  is directed toward the center of the circle and has constant magnitude  $a_{rad} = \frac{v^2}{R}$

## Period

The period  $T$  is the time required for the object to make one complete circle; the magnitude of the acceleration vector can also be written as

$$a_{rad} = \frac{4\pi^2 R}{T^2} \quad (44)$$

## Dynamics of Circular Motion(Cont.)

Provided that the object can be treated as a particle, circular motion (like other motions) is governed by Newton's second law,  $\sum \vec{F} = m\vec{a}$ .

### Centripetal Force in circular motion

Because the net force vector points in the same direction as the acceleration vector, an object in uniform circular motion is acted upon by a net force directed toward the center of the circle, with magnitude

$$F_{net} = m \frac{v^2}{R} \quad (45)$$

# Newton's Law of Gravitation

Two particles with masses  $m_1$  and  $m_2$  a distance  $r$  apart, attract each other gravitationally with forces of magnitude

$$F_g = G \frac{m_1 m_2}{r^2} \quad (46)$$

Where  $G = 6.67259 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$ , is the gravitational constant.

These forces form an action–reaction pair in accordance with Newton's third law.

Even if the objects cannot be treated as particles, but are spherically symmetric, this law is still valid; then,  $r$  is the distance between their centers of gravity.

## Newton's Law of Gravitation (Cont.)

The gravitational interaction of a spherically symmetric object, from any point outside the object, is just the same as though all its mass were concentrated at the object's center.

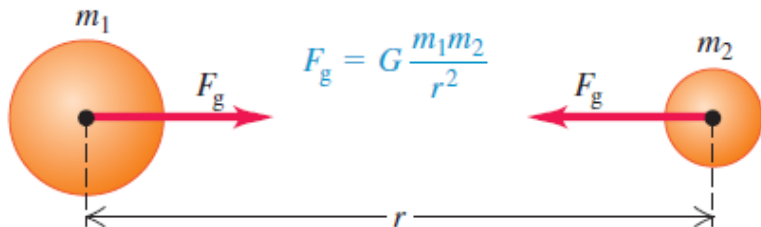


Figure 3: Gravitational attraction of two masses

# Weight

## Definition of the weight of a body

The weight  $W$  of a body is the total gravitational force exerted on it by all other bodies in the universe.

Near the surface of the earth, an object's weight is very nearly equal to the gravitational force exerted by the earth alone.

If the earth's mass is  $M_E$ , the weight of an object of mass  $m$  is

$$W = F_g = G \frac{mM_E}{R_E^2} \quad (47)$$

and the acceleration due to gravity  $g$  is

$$g = G \frac{M_E}{R_E^2} \quad (48)$$

## Satellites' Motion

When a satellite moves in a circular orbit, the centripetal acceleration is provided by the gravitational attraction of the astronomical body it orbits.

If the satellite orbits the earth in an orbit of radius  $r$ , its speed and period  $T$  are, respectively,

$$v = \sqrt{\frac{GM_E}{r}} \quad (49)$$

and,

$$T = \frac{2\pi r}{v} = 2\pi r \sqrt{\frac{r}{GM_E}} = \frac{2\pi r^{3/2}}{\sqrt{GM_E}} \quad (50)$$

# Satellites' Motion: Escape Speed

If an object is projected upward from Earth's surface with a large enough speed, it can soar off into space and never return.

This speed is called *Earth's escape speed (or the Second cosmic velocity)* and is given by

$$v_{\text{esc}} = v_2 = \sqrt{\frac{2GM_E}{R_E}} \quad (51)$$

The first and the third cosmic velocities are, respectively, given by

$$v_1 = \sqrt{\frac{GM_E}{R_E}} \quad (\text{The speed of the satellite around the Earth}) \quad (52a)$$

$$v_3 = \sqrt{\frac{2GM_S}{R_S}} \quad (\text{The speed for an object to escape the Solar System}) \quad (52b)$$

# Satellites' Motion: Kepler's Laws

The German astronomer Johannes Kepler discovered that the orbit of Mars could be accurately described by an ellipse with the Sun at one focus.

He then generalized this analysis to include the motions of all planets.

The complete analysis is summarized in three statements known as Kepler's laws:

- All planets move in elliptical orbits with the Sun at one of the focal points.
- A line drawn from the Sun to any planet sweeps out equal areas in equal time intervals.
- The square of the orbital period of any planet is proportional to the cube of the average distance from the planet to the Sun.

# Laws of Motion (Examples)

## Example No. 9

- ① A worker with spikes on his shoes pulls with a constant horizontal force of magnitude 20 N on a box with mass 40 kg resting on the flat, frictionless surface of a frozen lake. What is the acceleration of the box?
- ② A big luxury car weighing 19600 N, traveling in the  $+x$  direction, makes a fast stop; the  $x$  component of the net force on it is -1500 N. what is its acceleration?
- ③ To improve the acoustics in auditorium, a sound reflector with mass of 200 kg is suspended by a chain from the ceiling. What is its weight? What force (magnitude and direction) does the chain exert on it? What is the tension in the chain? Assume that the mass of the chain itself is negligible.

# Energy

Energy is one of the most important unifying concepts in all of physical science. Its importance stems from the principle of conservation of energy.

Energy is exchanged and transformed during interactions of systems, but the total energy in a closed system is constant.

Conservation laws, including conservation of mass, energy, momentum, angular momentum, electric charge, and other quantities, play vital roles in every area of physics and provide unifying threads that run through the whole fabric of the subject.

## Conservation Law

Every conservation law says that the total amount of some physical quantity in every isolated system is constant.

## Energy (Cont.)

This section is concerned with mechanical energy of three types:

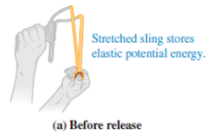
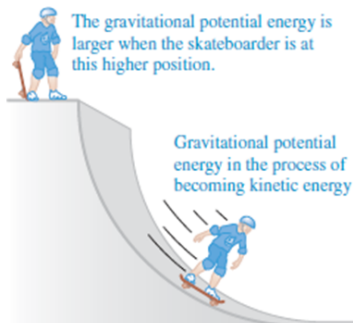
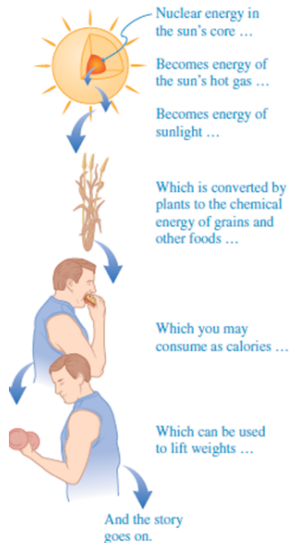
*Kinetic energy*, associated with the motion of objects that have mass;

*Gravitational potential energy*, associated with gravitational interactions;  
and

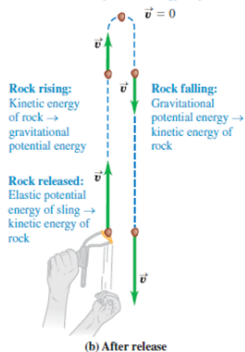
*Elastic potential energy*, associated with elastic deformations of objects.

Potential energy can be viewed as a stored quantity that represents the potential for doing work.

# Energy (Cont.)

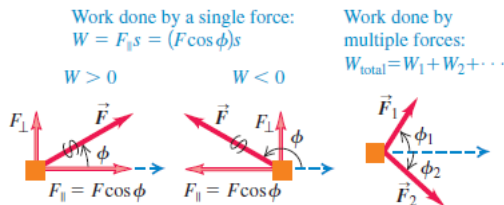


Top of trajectory:  
Gravitational potential energy only



# Work

When a force acts on an object that undergoes a displacement, the force does work on the object and transfers energy to it.



The work done by a constant force  $\vec{F}$  by displacing an object at  $\Delta \vec{x}$  is given by:

$$W = \vec{F} \cdot \Delta \vec{x} = F \Delta x \cos \theta \quad (53)$$

## Work (Cont.)

When an object moves under the action of a varying force parallel to the displacement, the work done by the force is represented graphically as the area under a graph of the force as a function of displacement.

If the magnitude of force  $F$  required to stretch a spring a distance  $x$  beyond its natural length is proportional to  $x$ , then

$$F = -kx \quad (54)$$

where  $k$  is a constant for the spring.

The total work needed to stretch the spring for  $X = 0$  to  $x = X$  is

$$W = \frac{1}{2}kX^2 \quad (55)$$

# Work and Kinetic Energy

## Kinetic Energy

The kinetic energy  $K$  of an object is defined in terms of the object's mass  $m$  and Speed  $v$  as

$$K = \frac{1}{2}mv^2 \quad (56)$$

## Work and Kinetic Energy

When forces act on an object, the change in its kinetic energy is equal to the total work done by all the forces acting on it:

$$W_{total} = K_f - K_i = \Delta K \quad (57)$$

When  $W$  is positive, the object speeds up; when  $W$  is negative, it slows down.

## Work and Potential Energy

Potential energy can be thought as stored energy. For an object of mass  $m$  at a height  $y$  above a chosen origin (where  $y = 0$ ), the gravitational potential energy is

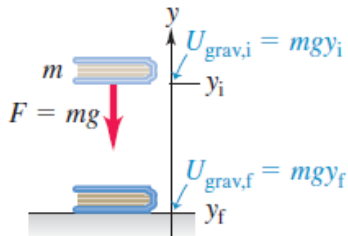
$$U_{grav} = mgy \quad (58)$$

When a spring is stretched or compressed a distance  $x$  from its uncompressed length, the spring stores elastic potential energy

$$U_{el} = \frac{1}{2}kx^2 \quad (59)$$

Potential energy is associated only with conservative forces, not with nonconservative forces such as friction.

# Work and Potential Energy (Cont.)

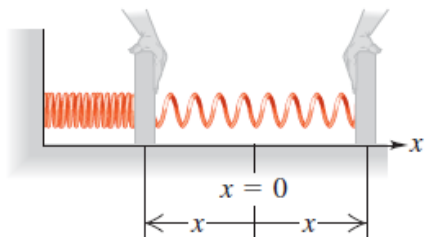


**Gravitational potential energy:**  
For an object at vertical position  $y$ ,

$$U_{\text{grav}} = mgy$$

For a change in vertical position,

$$\Delta U_{\text{grav}} = U_{\text{grav},f} - U_{\text{grav},i}$$



**Elastic potential energy:** For a spring stretched or compressed by a distance  $x$  from equilibrium,

$$U_{\text{el}} = \frac{1}{2} kx^2$$

# Conservation of Energy

When only conservative forces act on an object, the total mechanical energy (kinetic plus potential) is constant; that is,

$$K_i + U_i = K_f + U_f \quad (60)$$

If some of the forces are nonconservative, we label their work as  $W_{other}$ . The change in total energy (kinetic plus potential) of an object during any motion is equal to the work  $W_{other}$  done by the nonconservative forces:

$$W_{other} = \Delta KE + \Delta PE = (KE_f - KE_i) + (PE_f - PE_i) \quad (61)$$

or

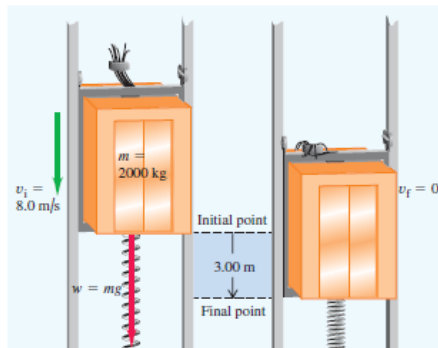
$$K_i + U_i + W_{other} = K_f + U_f \quad (62)$$

Nonconservative forces include friction forces, which usually act to decrease the total mechanical energy of a system.

# Laws of Motion (Examples)

## Example No. 10

In a worst-case design scenario, a 2000 kg elevator with broken cables is falling at 8.0 m/s when it first contact a cushioning spring at the bottom of the shaft. The spring is supposed to stop the elevator, compressing 3.0 m as it does so. Determine what the force constant of the spring should be.



# Power

Power is the time rate of doing work or the rate at which energy is transferred or transformed.

For mechanical systems, power is the time rate at which work is done by or on an object or a system.

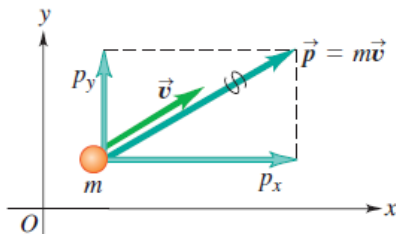
When an amount of work  $\Delta W$  is done during a time interval  $\Delta t$ , the average power is

$$P_{av} = \frac{\Delta W}{\Delta t} \quad (63)$$

# Definition of Linear Momentum

The momentum  $\vec{p}$  of a particle of mass  $m$  moving with velocity  $\vec{v}$  is defined as the vector quantity

$$\vec{p} = m\vec{v} \quad (64)$$



Momentum is a vector quantity; it has the same direction as the object's velocity and can be represented by components.

## Definition of Linear Momentum (Cont.)

Newton's second law can be restated as follows: "The rate of change of momentum of a particle equals the vector sum of the force acting on it":

$$\sum \vec{F} = \lim_{\Delta t \rightarrow 0} \frac{\Delta \vec{p}}{\Delta t} \quad (65)$$

The total momentum  $\vec{P}$  of a system is the vector sum of momenta of the individual particles:

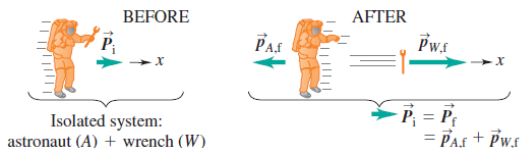
$$\vec{P} = \sum \vec{p}_i = m_i \vec{v}_i \quad (66)$$

# Conservation of Linear Momentum

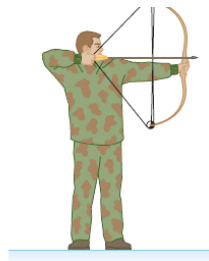
The total linear momentum of an isolated system is constant:

$$\vec{P}_i = \vec{P}_f \quad (67)$$

Each component of linear momentum is conserved separately.



(a)



(b)

**Figure 4:** (a) An Example of conservation of Linear Momentum. (b) An archer fires an arrow horizontally to the right. Because he is standing on frictionless ice, he will begin to slide to the left across the ice.

# Introduction

One of examples of the application of energy and momentum conservation laws in the study of the physical real process is the collision between bodies.

When two or more objects come close enough so that there is some sort of interaction between them, we say that *a collision takes place between the objects*. The interaction between colliding objects takes place during a very short time interval.



# Types of Collisions

Collisions can be classified according to energy relations and final velocities.

## Elastic Collision

In an elastic collision between two objects, kinetic energy is conserved and the initial and final relative velocities have the same magnitude.

## Inelastic Collision

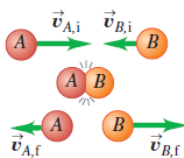
In an inelastic two-object collision, the final kinetic energy is less than the initial kinetic energy; if the two objects have the same final velocity (for instance, if they stick together), the collision is completely inelastic.

# Coefficient of restitution

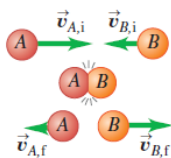
A coefficient known as *Coefficient of restitution* is used to classify collisions:

$$e = \frac{v_{B,f} - v_{A,f}}{v_{A,i} - v_{B,i}} \quad (68)$$

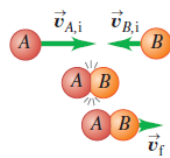
For a perfectly elastic collisions,  $e = 1$ . For inelastic collisions  $e < 1$ . For the bodies stick together after collision (perfectly inelastic collisions),  $e = 0$ .



**Elastic:**  
 $K$  conserved



**Inelastic:**  
Some  $K$  lost



**Completely inelastic:**  
Objects have same  $\vec{v}_f$

## Velocities before and after collision

In the case of the one dimensional collision, the velocities of two bodies of masses  $m_1$  and  $m_2$  after a perfectly elastic head-on collision are given by:

$$v_{1f} = \frac{(m_1 - m_2)v_{1i} + 2m_2v_{2i}}{m_1 + m_2} \quad (69a)$$

$$v_{2f} = \frac{(m_2 - m_1)v_{2i} + 2m_1v_{1i}}{m_1 + m_2} \quad (69b)$$

Velocity of motion of bodies after a perfectly inelastic head-on collision is given by:

$$\vec{v}_f = \frac{m_1\vec{v}_{1i} + m_2\vec{v}_{2i}}{m_1 + m_2} \quad (70)$$

## Velocities before and after collision (Cont.)

In two (or three dimensions), the concepts of conservation of momentum of the system implies that the momentum in each of the direction  $x$ ,  $y$ , and  $z$  is conserved.

For a 2D collisions, we obtain two component equations for conservation of momentum:

$$m_1 v_{1,i,x} + m_2 v_{2,i,x} = m_1 v_{1,f,x} + m_2 v_{2,f,x} \quad (71a)$$

$$m_1 v_{1,i,y} + m_2 v_{2,i,y} = m_1 v_{1,f,y} + m_2 v_{2,f,y} \quad (71b)$$

If the collision is elastic, we can also use the conservation of kinetic energy equation to get:

$$\frac{1}{2} m_1 v_{1,i}^2 + \frac{1}{2} m_2 v_{2,i}^2 = \frac{1}{2} m_1 v_{1,f}^2 + \frac{1}{2} m_2 v_{2,f}^2 \quad (72)$$

If the collision is inelastic, kinetic energy is not conserved and Equation (72) does not apply.

# Impulse

When a force acts on an object, the object's change in momentum depends on the force and on the time interval during which it acts.

The impulse  $\vec{J}$  of a constant force  $\vec{F}$  acting over a time interval  $\Delta t$  from  $t_i$  to  $t_f$  is the vector quantity

$$\vec{J} = \vec{F}(t_f - t_i) = \vec{F}\Delta t \quad (73)$$

The change in momentum of a particle during any time interval equals the impulse of the total force acting on the particle:

$$\Delta \vec{P} = \vec{F}\Delta t = \vec{J} \quad (74)$$

# Collision (Examples)

## Example No. 11

- 1 A marksman holds a 3.00 kg rifle loosely, allowing it to recoil freely when fired, and fires a bullet of mass 5.00 g horizontally with speed of 300 m/s. What is the recoil speed of the rifle? What are the final kinetic energies of the bullet and the rifle
- 2 A small compact car with a mass of 1000 kg is traveling north on Morewood avenue with a speed 15 m/s. At the intersection of the Morewood and Fifth avenue, it collides with a truck with a mass 2000 kg that is traveling east of the Fifth avenue at 10 m/s. Treating each vehicle as a particle, find the total momentum (magnitude and direction) just before the collision.

## Collision (Examples (Cont.))

### Example No. 11 (Cont.)

3. In previous question (2), we consider the impending collision of small car with a larger truck. It's now 2 seconds later, and the collision has occurred. Fortunately, all occupants were wearing seat belts, and there were no injuries; but the two vehicles became thoroughly tangled and moved away from the point of impact as one mass. The insurance adjuster has asked you to help find the velocity of the wreckage just after impact.

4. Suppose you throw a ball with mass  $0.40 \text{ kg}$  against a brick wall. It hits the wall moving horizontally to the left at  $30 \text{ m/s}$  and rebounds horizontally to the right at  $20 \text{ m/s}$ . Find the impulse of the force exerted on the ball by the wall. If the ball is in contact with the wall for  $0.010 \text{ s}$ , find the average force on the ball during the impact.

# The Centre of the mass of a system

The mass of a system of particles is the sum of masses of all particles of that system

$$m = \sum_{i=1}^n m_i \quad (75)$$

The coordinates of the centre of the mass of a system are defined as:

$$x_c = \frac{\sum m_i x_i}{\sum m_i}, \quad y_c = \frac{\sum m_i y_i}{\sum m_i} \text{ and } z_c = \frac{\sum m_i z_i}{\sum m_i} \quad (76)$$

Thus, the position-vector of the centre of mass of a system is given by

$$\vec{r}_c = \frac{\sum m_i \vec{r}_i}{\sum m_i} \quad (77)$$

## The Centre of the mass of a system (Cont.)

The center of mass of a system moves as though all the mass were concentrated at the center of mass.

The equation of motion of the centre of mass is

$$m \frac{d\vec{v}_c}{dt} = \sum \vec{F}_i \quad (78)$$

where  $m$  is the whole mass of the system, and  $\vec{v}_c$  is the velocity of the centre of the mass.

## The Centre of the mass of a system (Cont.)

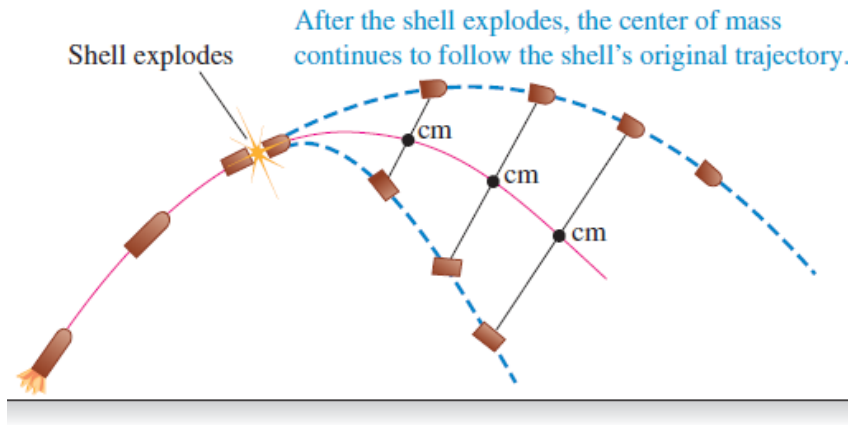


Figure 5: *A projectile explodes in flight*

# Outline

## 5 ROTATION OF RIGID BODIES

- Introduction
- Rotational Kinetic Energy and Inertia
- Torque
- Angular Momentum
- Stability

When an extended object such as a wheel rotates about its axis, the motion cannot be analyzed by modeling the object as a particle because at any given time different parts of the object have different linear velocities and linear accelerations. In this section, we analyze the motion of an extended object by modeling it as a system of many particles, each of which has its own linear velocity and linear acceleration.

Having successfully completed this section, you should be able to:

-  Define a rigid body;
-  Make a distinction between the translational and the rotational motion;
-  Describe the rotational motion of rigid bodies using the concepts of angular velocity and acceleration, rotational inertia, torque, and the rotational analog of Newton's law;
-  Calculate the rotational kinetic energy, and explain its relation to torque and work;
-  Discuss the law of conservation of the angular momentum and apply it to given situations.

# A Rigid Body

A rigid body is defined as a system consisting of a large number of point masses called particles, such that the distances between the pairs of masses remain constant even when the body is in motion or under the action of external forces.

This is an idealized definition of a rigid body. This is because in nature, there is no such thing as true point masses or particles and no body of any physical size is strictly rigid; it deforms under the action of applied forces.

# Rotational Kinetic Energy

Just like massive bodies tend to resist changes in their motion. Rotating bodies also tend to resist changes in their motion. We call this *rotational inertia*.

We can determine its expression by looking at kinetic energy.

$$\begin{aligned} K &= \frac{1}{2}mv^2; \quad \text{but} \quad v = r\omega; \\ &= \frac{1}{2}m(r\omega)^2 \\ &= \frac{1}{2}mr^2\omega^2; \quad \text{and defining} \quad mr^2 = I; \end{aligned}$$

Thus the rotational kinetic energy is given by

$$K = \frac{1}{2}I\omega^2 \quad (79)$$

## Moment of inertia

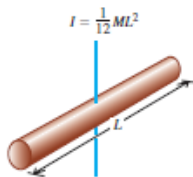
The moment of inertia of a system (a body) about a given axis is a physical quantity given by

$$I = \sum m_i \vec{r}_i^2 \quad (80)$$

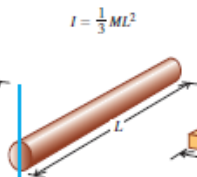
When the mass of the body is uniformly (continuously) distributed, the sum is reduced to integral as:

$$I = \int_V r^2 dm \quad (81)$$

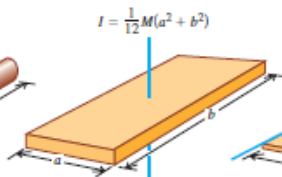
# Moments of Inertia for various bodies



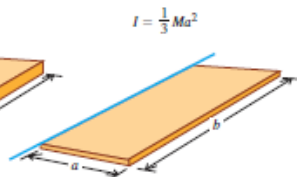
(a) Slender rod,  
axis through center



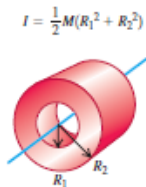
(b) Slender rod,  
axis through one end



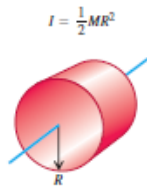
(c) Rectangular plate,  
axis through center



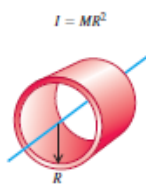
(d) Thin rectangular plate,  
axis along edge



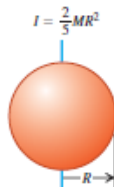
(e) Hollow cylinder



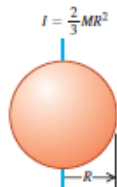
(f) Solid cylinder



(g) Thin-walled hollow  
cylinder



(h) Solid sphere



(i) Thin-walled hollow  
sphere

## Steiner's Theorem

The moment of inertia  $I$  of a body about an arbitrary axis is given by:

$$I = I_c + Ma^2 \quad (82)$$

where  $I_c$  is the moment of inertia about an axis through the centre of the mass,  $M$  the total mass of the system and,  $a$  is the distance of the arbitrary axis (for which  $I$  is calculated) to the first axis.

The kinetic energy of a body rolling on inclined plane without sliding is given by

$$T = \frac{1}{2}mv_c^2 + \frac{1}{2}I_c\omega^2 \quad (83)$$

# MOI (Examples)

## Example No. 12

Consider two equal masses (3 kg) at the end of a bar (with negligible mass) of 4 m long rotating about a specified axis. Calculate:

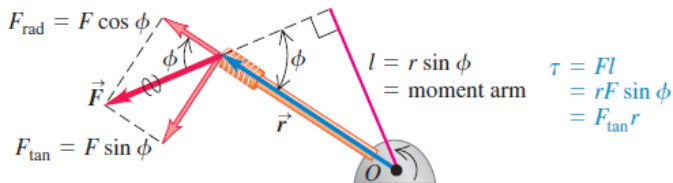
- 1 The moment of inertia when they are rotating around the center of their rod.
- 2 The moment of inertia when they are rotating at the end of the rod.
- 3 The moment of inertia when they are rotating at a point 2 m from one end of the rod.
- 4 Rework (2) and (3) using Steiner's theorem.

# Definition

When a force  $\vec{F}$  acts on a body, the torque  $\vec{\tau}$  of that force with respect to a point  $O$  is given by

$$\vec{\tau} = \vec{r} \times \vec{F} \quad (84)$$

We use the sign convention that a torque tending to cause counterclockwise rotation is positive, while one tending to cause clockwise rotation is negative.

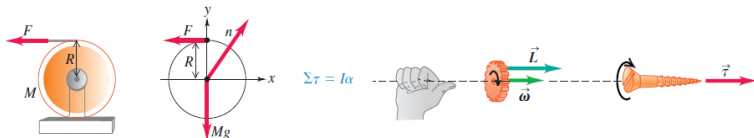


# Torque and Angular Acceleration

The angular acceleration  $\alpha$  of a rigid body rotating about a stationary axis is related to the torque and the moment of inertia of the body by

$$\sum \tau = I\alpha \quad (85)$$

This relation is also valid for a body with a moving axis of rotation, provided that the rotation axis is an axis of symmetry that passes through the center of mass and does not change direction.



## Work done by the Torque

When a constant torque acts on a rigid body that undergoes an angular displacement  $\Delta\theta$  from  $\theta_1$  to  $\theta_2$ , the work  $W$  done by the torque is given by

$$W = \tau(\theta_2 - \theta_1) = \tau\Delta\theta \quad (86)$$

When the object rotates with angular velocity  $\omega$ , the power (rate at which the torque does work) is

$$P = \tau\omega \quad (87)$$

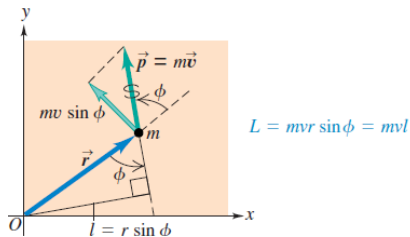
# Definition

The angular momentum  $\vec{L}$ , with respect to any axis through a point  $O$ , of a particle with mass  $m$  and velocity  $\vec{v}$  is a physical quantity defined by

$$\vec{L} = \vec{r} \times m\vec{v} \quad (88)$$

When a rigid body with moment of inertia  $I$  rotates with angular velocity  $\vec{\omega}$  about a stationary axis, the angular momentum with respect to that axis is

$$\vec{L} = I\vec{\omega} \quad (89)$$



# Angular Momentum and Torque

In term of torque and angular momentum , the basic dynamic relation for rotation motion can be restated as

$$\sum \vec{\tau} = \frac{\Delta \vec{L}}{\Delta t} \quad (90)$$

If a system consists of bodies that interact with each other but not with anything else, or if the total torque associated with the external forces is zero, then the total angular momentum of the system is constant (or conserved).

$$\Sigma \tau = \frac{d\vec{L}}{dt} = 0, \quad \vec{L} = 0 \quad (91)$$

This principle is universal conservation law, valid at all scales from atomic and nuclear systems to the motion of galaxies.

# Equilibrium of Rigid Body

The two conditions for equilibrium of a rigid body are

$$\sum \vec{F} = 0 \quad (92a)$$

$$\sum \vec{\tau} = 0 \quad (92b)$$

The former ensures that the center of mass of the body has no linear acceleration  $a$ , and the latter ensures that it has no angular acceleration  $\alpha$  about any axis.

# Overview

If a body is disturbed from equilibrium, it generally experiences nonzero torques or forces that cause it to accelerate.

Normally, there are four different types of equilibrium: *stable*, *unstable*, *neutrally stable*, and *metastable*.

The lowest point in a valley is stable.



The highest point on a hill is unstable.



A level surface is neutrally stable.

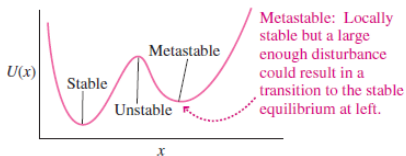


This point is metastable.



Note that the hill goes lower over here.

(a)



(b)

In terms of potential energy, we have:

$$\frac{dU}{dx} = 0 \quad (\text{equilibrium condition}) \quad (93a)$$

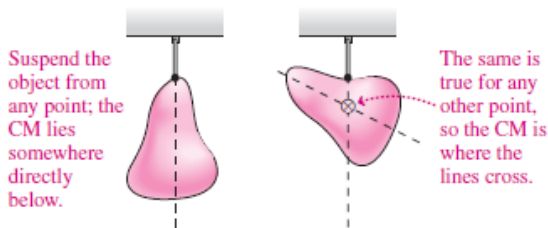
$$\frac{d^2U}{dx^2} > 0 \quad (\text{stable equilibrium condition}) \quad (93b)$$

$$\frac{d^2U}{dx^2} < 0 \quad (\text{unstable equilibrium condition}) \quad (93c)$$

# Centre of Gravity

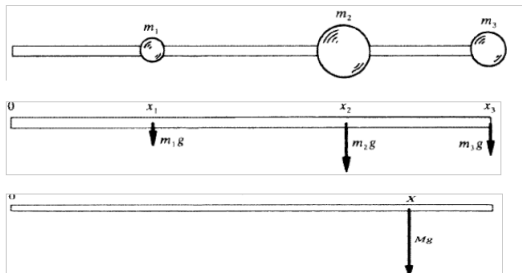
The center of gravity of a system is the point where the force of gravity appears to act. When the gravitational field is uniform over the system, the center of gravity coincides with the center of mass.

The following figure provides a handy way to locate the center of mass.



## Centre of Gravity (Cont.)

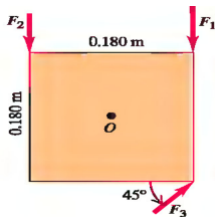
In the case of a complex object, the way to find its center of gravity is to consider it as a system of separate particles and then find the balance point of the system.



$$X = \frac{m_1x_1 + m_2x_2 + m_3x_3}{m_1 + m_2 + m_3} = \frac{\sum_i m_i x_i}{\sum_i m_i} \quad (94)$$

## Example No. 13

- ① A square metal  $0.180\text{ m}$  on each side is pivoted about an axis through point  $O$  at its center and perpendicular to the plate. Calculate (a) The torque (magnitude and direction) about this axis due to each of three forces shown in the figure, if the magnitude of the forces are  $F_1 = 18.0\text{ N}$ ,  $F_2 = 26.0\text{ N}$  and  $F_3 = 14.0\text{ N}$ . (b) The net torque about this axis due to the three forces.



- ② Where must a  $0.80\text{-kN}$  object be hung on a uniform, horizontal, rigid  $100\text{-N}$  pole so that a girl pushing at one end supports one-third as much as a woman pushing up at the other end?

# Outline

## 6 ELASTIC PROPERTIES OF SOLIDS

Although we often talk about rigid bodies there are no absolute rigid bodies, for every body alters its size or shape under the action of applied force. Elasticity is the property of a material body by virtue of which it recovers its original shape or size on removal of the external deforming forces. In this section, we discuss the concept of elasticity.

Having successfully completed this section, you should be able to:

-  Explain the fundamental laws of elasticity;
-  Describe different moduli of elasticity;
-  Describe the stress–strain curve and to define yield point, elastic limit, permanent set, and brittle and ductile materials;
-  Explain elastic potential energy of a deforming body in terms of stress and strain.

# Elasticity

If the distance between any two points in a body is invariable, the body is said to be a ***rigid body***.

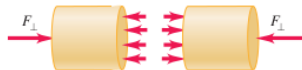
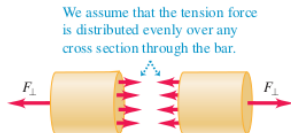
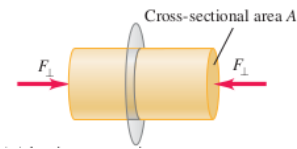
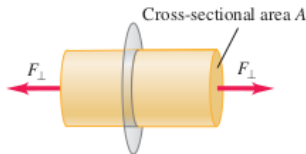
In practice, it is not possible to have a perfect rigid body. All bodies get deformed under the action of force-some more some less.

The property of the body by virtue of which it tends to regain its original geometry (shape or size) on the removal of deforming forces is called elasticity.

If a body resists deformation and recovers its original shape, size or volume completely and immediately on the removal of the deforming forces it is called perfectly elastic body and if it completely retains its altered size and shape, it is said to be perfectly plastic body.

# Tensile and Compressive Stress and of Strain

$$\text{Tensile (compressive) Stress} \equiv \sigma = \frac{F_{\perp}}{A} \quad (95)$$



$$\text{Tensile (compressive) Strain} \equiv \epsilon = \frac{l - l_o}{l_o} = \frac{\Delta l}{l_o} \quad (96)$$

# Young's modulus

Experiments have shown that, for a sufficiently small tensile or compressive stress, stress and strain are proportional, as stated by Hook's law, 43. The corresponding proportionality constant, called **Young's modulus** is given by

$$Y = \frac{\text{Tensile stress}}{\text{Tensile strain}} \quad \text{or} \quad \frac{\text{Compressive stress}}{\text{Compressive strain}}$$

or

$$Y = \frac{F_{\perp}/A}{\Delta l/l_o} = \frac{l_o}{A} \frac{F_{\perp}}{\Delta l} \quad (97)$$

Young's modulus is a property of a specific material, rather than of any particular object made of that material.

# Volume Stress and Strain and Bulk Modulus

When a body is under water, the water exerts nearly uniform pressure everywhere on its surface and squeezes it to a slightly smaller volume. This situation is different from tensile and compressive stresses and strains. The stress is now a uniform pressure on all sides, and the resulting deformation is a change in volume.

The **pressure** in a fluid, denoted by  $p$ , is the force  $F_{\perp}$  per unit area  $A$  transmitted across any cross section of the fluid, against a wall of its container, or against a surface of an immersed object:

$$p = \frac{F_{\perp}}{A} \quad (98)$$

The corresponding strain is given by

$$\text{Volume strain} = \frac{\Delta V}{V_o} \quad (99)$$

## Volume Stress and Strain and Bulk Modulus (Contd)

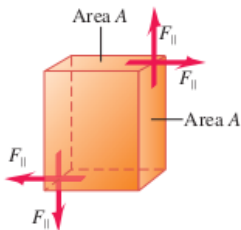
When Hooke's law is obeyed, the volume strain is proportional to the volume stress (change in pressure).

The corresponding constant ratio of stress to strain is called the **bulk modulus**, denoted by  $B$ . When the pressure on an object changes by a small amount  $\Delta p$ , from  $p_o$  to  $p_o + \Delta p$ , and the resulting volume strain is  $\Delta V/V_o$ , Hooke's law takes the form

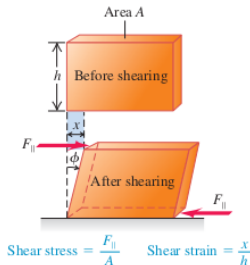
$$B = -\frac{\Delta p}{\Delta V/V_o} \quad (100)$$

We include a minus sign in this equation because an increase in pressure always causes a decrease in volume.

# Shear Stress and Strain



A block subjected to shear stress from two directions



If the forces are small enough so that Hooke's law is obeyed, the shear strain is proportional to the shear stress.

The corresponding proportionality constant is called the **shear modulus**, denoted by  $S$ :

$$S = \frac{\text{Shear stress}}{\text{Shear strain}} = \frac{F_{\parallel}/A}{x/h} = \frac{h}{A} \frac{F_{\parallel}}{x} = \frac{F_{\parallel}/A}{\phi} \quad (101)$$

## Poisson's Ratio

When the deformation of a cylindrical body is in all directions, two strains are produced. Strain along the direction of the force (***longitudinal strain***) and strain in a perpendicular direction (***lateral strain***).

The ***Poisson's ratio*** or ***Poisson coefficient*** is defined as the ratio of fractional change of thickness  $\Delta d/d$  and the fractional change of length  $\Delta L/L$ .

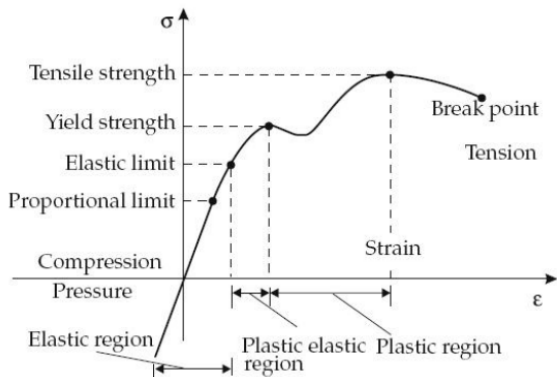
$$\mu = \frac{\text{lateral strain}}{\text{longitudinal strain}} = -\frac{\Delta d/d}{\Delta L/L} \quad (102)$$

The negative sign expresses the experimental fact the diameter  $d$  of a cylindrical wire is reduced under tension while its length  $L$  increases.

# Stress-Strain Diagram

The relationship between the stress and strain that a particular material displays is known as that material's ***Stress-Strain curve*** or ***Stress-strain diagram***.

It is unique for each material



## Work done in a Strain

In an elastic deformation of a body, work is performed. This work done by the applied force remains stored up in the body in the form of potential energy.

$$Y = \frac{F/A}{\Delta l/L}, \quad \text{hence} \quad F = \frac{Y A l}{L}$$

$$dW = F dl = \frac{Y A}{L} l dl$$

$$W = \int dW = \frac{Y A}{L} \int l dl = \frac{1}{2} \frac{Y A l^2}{L} = \frac{1}{2} F l \quad (103)$$

The work done per unit volume of the wire or the strain energy per unit volume is

$$w = \frac{W}{V} = \frac{1}{2} \frac{F l}{A L} = \frac{1}{2} \text{stress} \times \text{strain} \quad (104)$$

# Outline

## 7 FLUID MECHANICS

- Introduction
- Properties of Fluids
- Hydrostatics
- Hydrodynamics

A tornado whirls across a darkened sky. A plane flies, supported by air pressure on its wings. Fluid in a car's brake system amplifies the force of the foot on the brake pedal. Our bodies are sustained by air moving in and out our lungs, and by the flow of blood throughout our tissues. All these examples involve fluid motion. In our treatment of mechanics of fluids, we will be applying principles and analysis models that have already discussed. First, we will consider the mechanics of a fluid at rest, that is **fluid statics**, and then study fluids in motion, that is **fluid dynamics**.

Having successfully completed this section, you should be able to:

-  Define and describe properties of fluids;
-  Explain the surface tension and related phenomenon of capillarity, and apply them in everyday life;
-  Explain the relation between pressure and force;
-  Explain Pascal's law and Archimedes' principles;
-  Apply Pascal's law and Archimedes's principle in everyday life;
-  Express conservation of mass and energy for fluids through the continuity equation and Bernoulli's equation, and use the two to solve engineering problems involving fluid dynamics.

# States of Matter

## Solid

The atoms in a solid are held at specific positions with respect to one another by forces that are mainly electrical in origin. The atoms of solid vibrate about these equilibrium positions because of thermal agitation.

## Liquid

Thermal agitation is greater in the liquid state than in the solid state. As a result, the molecular forces in a liquid are not strong enough to keep the molecules in fixed positions, and the molecules wander through the liquid in random fashion

## Gas

In gaseous state, the molecules are in constant random motion and exert only weak forces on each other. The average separation distances between the molecules of the gas are quite large compared with the dimensions of the molecules.

# Density

## Fluid

A fluid is a substance which flows when an external force is applied on it. Fluids do not maintain a fixed shape, they have the ability to flow.

## Density

The density of a substance is the quantity of matter contained in a unit volume of the substance. It can be expressed in three different ways.

- Mass Density,  $\rho = \frac{m}{V}$
- Weight Density,  $\omega = \frac{W}{V} = \frac{mg}{V} = \rho g$
- Specific Gravity or Relative density,  $d = \frac{\text{mass density of Substance}}{\text{Mass density of Water at } 4^{\circ}\text{C}}$

# Viscosity

The viscosity of a fluid is that property which determines the amount of its resistance to a shearing force. Viscosity is due primarily to interaction between molecules.

## Dynamic Viscosity

Coefficient of dynamic viscosity (or absolute viscosity),  $\mu$ , is given by

$$\tau = \frac{F}{A} = \mu \frac{dv}{dy} \quad (105)$$

where,  $\tau$  is the shear stress,  $dv$  the variation of the velocity, and  $dy$  the variation of the layer of the fluid.

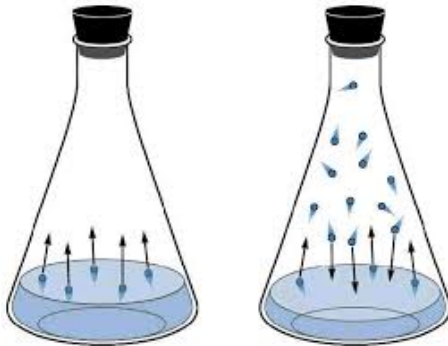
## Kinematic Viscosity

The coefficient of kinematic viscosity is given by

$$\nu = \frac{\mu}{\rho} \quad (106)$$

# Vapor Pressure

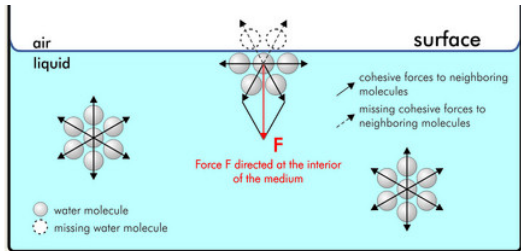
When evaporation takes place within an enclosed space, the partial pressure created by the vapor molecules is called vapor pressure.



Vapor pressure depend upon temperature and increase with it.

# Surface tension

The surface tension is the property of the liquid by virtue of which the free surface of liquid at rest tends to have minimum area and as such it behaves as if covered with a stretched membrane.



# Surface tension (Cont.)

## Surface Tension

The surface tension of a liquid is measured as the force acting on unit length of a line imagined to be drawn tangentially anywhere on the free surface of the liquid at rest. It acts at right angles to this line on both the sides and a long tangent to the liquid surface.

$$\sigma = \frac{F}{L} \quad (107)$$

## Surface Energy

The surface energy is defined as the amount of work done against the force of surface tension in forming the liquid surface of a given area at a constant temperature.

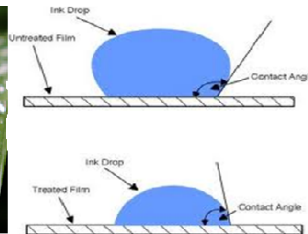
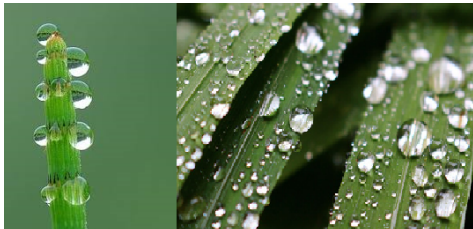
$$W = \sigma \cdot \Delta A \quad (108)$$

With  $\sigma$ , the surface tension and  $\Delta A$ , the increase in area.

# Capillarity

## Adhesion and Cohesion properties

Rise or fall of liquid in a capillary tube (or in porous media) is caused by surface tension and depends on the relative magnitudes of the cohesion of the liquid and the adhesion of the liquid to the walls of the containing vessel. Liquids rise in tubes they wet ( $\text{adhesion} > \text{cohesion}$ ) and fall in tubes they do not wet ( $\text{adhesion} < \text{cohesion}$ ).

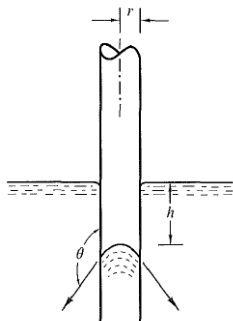
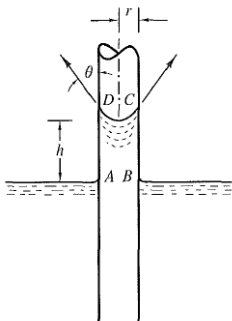


# Capillarity (Cont.)

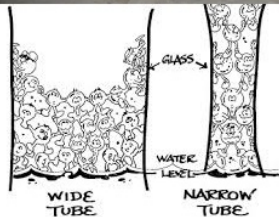
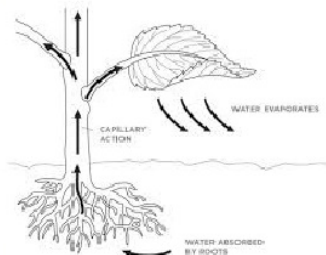
## Capillarity

The capillary rise (or depression) in a tube of the radius  $r$  is given approximately by

$$h = \frac{2\sigma \cos \theta}{\omega r} \quad (109)$$



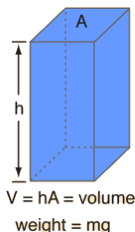
# Capillarity (Cont.)



# Pressure

Pressure is defined as force per unit area, where the force  $F$  is acting perpendicularly to the surface  $A$ .

$$p = \frac{F}{A} \quad (110)$$



Static fluid pressure does not depend on the shape, total mass, or surface area of the liquid.

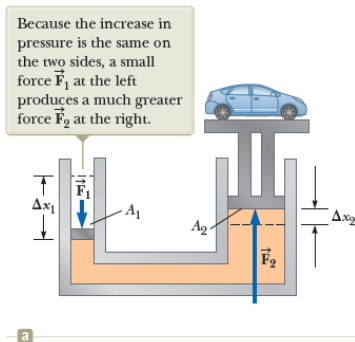
$$\text{Pressure} = \frac{\text{weight}}{\text{area}} = \frac{mg}{A} = \frac{\rho Vg}{A} = \rho gh$$



☛ The pressure is measured using a **manometer** or a **barometer**.

# Pascal's law (or Principle)

A pressure change occurring anywhere in a confined incompressible fluid is transmitted throughout the fluid such that the same change occurs everywhere.



**Figure 6:** One of applications of Pascal's law (a) *Diagram of a hydraulic press.* (b) *A vehicle undergoing repair is supported by a hydraulic lift in a garage.*

# Buoyancy and Archimedes' Principle

Buoyancy is a familiar phenomenon: An object immersed in water seems to weigh less than when it is in air.

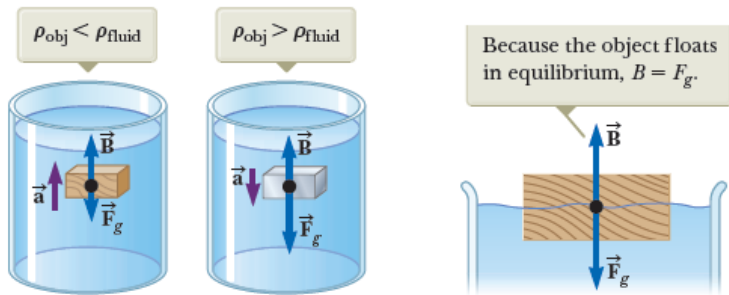
When a body is partially or wholly immersed in a fluid, it displaces the fluid.

The displaced fluid has a tendency to regain its original position. Due to this an upwards force is exerted on the body by the displaced fluid. This upwards force is called the Thrust or the Buoyant force.

## The Archimedes's principle

***When an object is completely or partially immersed in a fluid, the fluid exerts an upward force on the object equal to the weight of the fluid that is displaced by the object.***

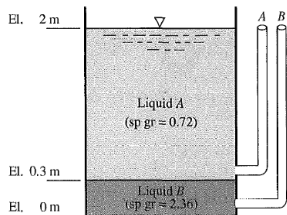
# Buoyancy (Cont.)



# Hydrostatics (Examples)

## Example No. 13

- ① For the open tank, with piezometers attached on the side, containing two different immiscible liquids as shown in the figure, find (a) the elevation of the liquid surface in piezometer A, (b) the elevation of the liquid surface in piezometer B, and (c) the total pressure at the bottom of the tank.



- ② A stone weighs 90 N in air, and when immersed in water it weighs 50 N. Compute the volume of the stone and its specific gravity.

# Characteristics of flow

We turn now from the study of fluids at rest to the more complex subject of fluid in motion, which is called hydrodynamics.

We can distinguish two main types of fluid flow:

## Laminar flow

If the flow is smooth such that neighboring layers of the fluid slide by each other smoothly, the flow is said to be *streamline or laminar flow*.

This kind of flow is characterized by the fact that each particle of the fluid follows a smooth path, and these paths do not cross over one another.

## Turbulent flow

Above a certain speed, which depends on a number of factors as we shall see later, the flow becomes *turbulent*.

Turbulent flow is characterized by erratic, small whirlpool-like circles called eddy currents or eddies.

# Reynolds' Number

Reynolds number is a dimensionless quantity that is used to determine the type of flow pattern as laminar or turbulent while flowing through a pipe.

The Reynolds' number is given by:

$$Re = \frac{D\rho v}{\mu} \quad (111)$$

Where:

$Re$  = reynolds number

$\mu$  = dynamic viscosity of the fluid

$\rho$  = density of the fluid

$v$  = flow speed

$D$  = characteristic linear dimension

# Reynolds' Number

To classify whether the flow is Turbulent or Laminar, an index known as *Reynolds' number* is used. The Reynolds' number is given by:

$$Re = \frac{D\rho v}{\mu} \quad (112)$$

For the pipe flow, we have:

- when  $Re < 2000$ , the flow is laminar;
- when  $Re > 4000$ , the flow is turbulent, and
- when  $2000 < Re < 4000$ , the flow may be either laminar or turbulent, depending on how insulated the flow is from outside disturbances.

For the open channel flow

- when  $Re < 500$ , the flow is laminar;
- when  $Re > 2000$ , the flow is turbulent, and
- when  $500 < Re < 2000$ , the flow is transitional.

# Flow rate and the equation of continuity

The flow rate is defined as the quantity of the flowing fluid by unit of time. The equation of continuity expresses conservation of mass for an incompressible fluid flowing in a tube or some other channel.

## Equation of Continuity

*The amount (either mass or volume) of fluid flowing through a cross section of the tube in a given time interval must be the same for all cross sections.*

or

$$Q = \frac{\Delta m}{\Delta t} = \text{constant} \quad \text{but} \quad \Delta m = \rho \times \Delta V$$

$$= \rho \frac{\Delta V}{\Delta t} = \rho A \frac{\Delta x}{\Delta t} = \rho A v$$

or

$$\rho A v = \text{Constant}$$

# Bernoulli's Principle

According to the continuity equation, the speed of fluid flow can vary along the paths of the fluid particles.

The pressure can also vary; it depends on height, as in the static situation, and it also depends on the speed of flow.

We can now use the concepts of work–energy theorem, to derive an important relationship called Bernoulli's equation that relates the pressure, flow speed, and height for flow in an ideal fluid.

## Bernoulli's Principle

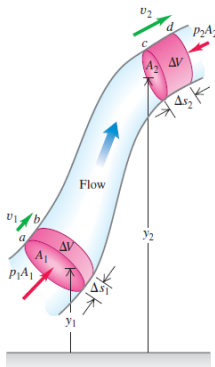
$$p + \frac{1}{2}\rho v^2 + \rho gy = \text{constant} \quad (113)$$

Bernoulli's principle states that an increase in the speed of a fluid occurs simultaneously with a decrease in static pressure or the fluid's potential energy.

# Bernoulli's Principle (Cont.)

Bernoulli's equation relates the pressure, flow speed and elevation in an ideal fluid. For any two points

$$p_1 + \frac{1}{2}\rho v_1^2 + \rho g y_1 = p_2 + \frac{1}{2}\rho v_2^2 + \rho g y_2 \quad (114)$$



# Applications of Bernoulli's Principle

## Torricelli's Theorem

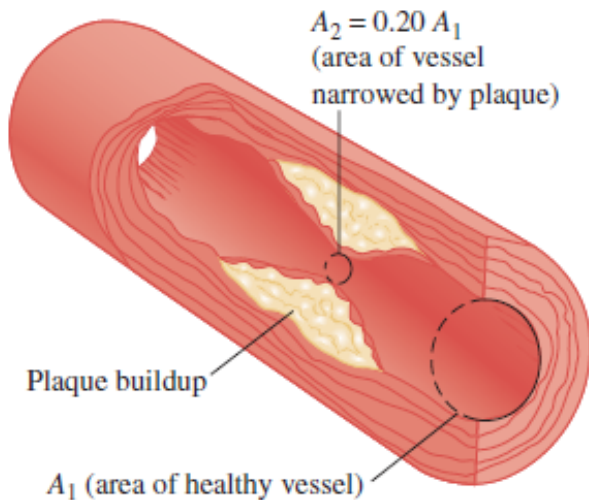
The Torricelli's Theorem gives us the velocity of the flow in an orifice perforated to the water vessel (tank):

$$v = \sqrt{2g(y_2 - y_1)} \quad (115)$$

## Collapse of a Blood Vessel

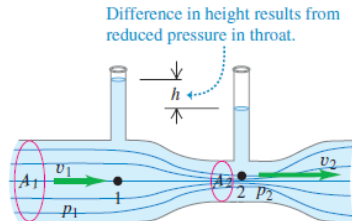
Over 200,000 people die each year from arteriosclerosis. One aspect of this disease is a narrowing of blood vessels due to the buildup of plaque on the vessel wall. The disease is considered critical when a portion of vessel becomes 80% blocked and a pressure drop inside the narrow region can act to collapse the blood vessel.

# Applications of Bernoulli's Principle (Cont.)



# Applications of Bernoulli's Principle (The Venturi Meter)

The Venturi meter, is used to measure flow speed of the fluid in a pipe.



$$v_1 = \sqrt{\frac{2gh}{(A_1/A_2)^2 - 1}} \quad (116)$$

This same concept is used in some gasoline engines, in which gasoline is sprayed from a nozzle or an injector into a low-pressure region produced in a Venturi throat.

## Laminar flow in Tubes-Poiseuille's Equation

If fluid has no viscosity, it could flow through a level pipe without a force being applied. Because of viscosity, a pressure difference between the ends of the tube is necessary for the steady flow of any real fluid, be water or oil in a pipe, or blood in the circulatory system of a human.

The rate of flow of a fluid in a round tube depends on the viscosity of the fluid, the pressure difference, and the dimensions of the tube.

The French scientist J.L. Poiseuille (1799-1869), who was interested in the physics of the blood circulation, determined how the variables affect the flow rate of an incompressible fluid undergoing laminar flow in a cylindrical tube.

$$Q = \frac{\pi R^4 (p_1 - p_2)}{8\eta L} \quad (117)$$

# Stoke's Law

The backwards dragging force  $F$ , acting on a small spherical body of radius  $r$ , moving through a medium of coefficient of viscosity  $\eta$ , with the velocity  $v$  is given by:

$$F = 6\pi\eta rv \quad (118)$$

- This law accounts the formation of clouds;
- It helps a man falling with the help of a parachute.

# Terminal Velocity

It is the maximum constant velocity acquired by the body which falling freely in a viscous medium. When a small spherical body falls freely through a viscous medium, three forces act on it.

- Weight,  $W = \frac{4}{3}\pi r^3 \rho_o g$
- Upwards thrust (buoyancy),  $F_T = \frac{4}{3}\pi r^3 \rho_f g$
- Viscous drag  $F_v = 6\pi\eta r v$

Applying Newton's second law, we get

$$v_t = \frac{2r^2(\rho_o - \rho_f)g}{9\eta} \quad (119)$$

# Hydrodynamics (Examples)

## Example No. 14

- 1 What must be the gauge pressure in a large fire hose if the nozzle is to shoot water straight upward to a height of 30.0 m?
- 2 A 14 cm inner diameter water main furnishes water (through intermediate pipes) to 1.0 cm inner diameter faucet pipe. If the average speed in the faucet is  $3.0 \text{ cm/s}$ , what will be the average speed it causes the water main?

# Outline

## 8 OSCILLATION AND WAVES

- Introduction
- Simple Harmonic Motion (SHM)
- Examples of SHMs
- Damped Oscillations
- Forced Oscillation
- Mechanical Waves

Ripples on a pond, music sounds, seismic tremors triggered by an earthquake, all these are wave phenomena. In this section, we are going to study oscillating motion and mechanical waves. We will start with simple harmonic motion (SHM), then damped and forced oscillations, and we close this study with the description of mechanical waves.

Having successfully completed this section. you should be able to:

-  Define and explain the harmonic motion;
-  Discuss the examples of simple harmonic motion in everyday life;
-  Explain how energy is shared between potential and kinetic energies for a body in simple harmonic motion;
-  Explain the nature of waves as traveling disturbances that transport energy but not matter;
-  Describe waves mathematically in terms of amplitude, period, frequency, and wavelength.
-  Calculate the rate at which a mechanical wave transport energy;
-  Understand the wave equation and use it to determine wave speeds.

## Definition

The vibration of a quartz crystal of modern electric watch, the swinging pendulum motion of grandfather watch, the sound vibrations produced by a clarinet or any organ pipe, the back-forth motion of the pistons in a car engine are all examples of repeated motions over and over, called "***periodic motions***" or "***oscillations***".

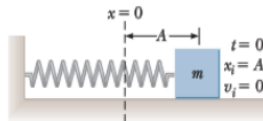
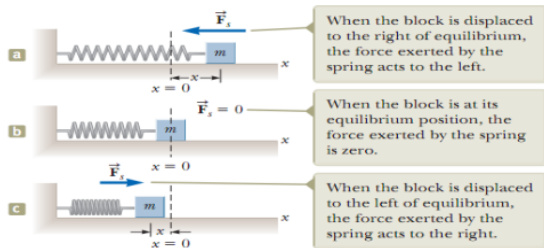
A period motion always has a stable equilibrium position. When a body in periodic motion is moved away from equilibrium and then released, a force  $\vec{F}$  or torque  $\vec{\tau}$  comes into action to pull it back toward equilibrium. Meanwhile it picks up some kinetic energy, so it overshoots, stopping somewhere on the other side, and again pulled back.

The spring vector force  $\vec{F}$  responsible of the oscillatory motion is called "a restoring force".

**An example** is the motion of a simple pendulum or the motion of an elastic pendulum.

# Motion of an Object Attached to a spring

One of the simplest systems having a periodic motion is a body of mass  $m$  attached on a spring and moving on a frictionless horizontal guide system (linear air track) along the  $\overline{OX}$  axis.



Using the Newton's second law, we get:

$$F = ma, \quad \text{with} \quad F = -kx$$

and

$$a = \frac{d^2x}{dt^2} \Rightarrow m \frac{d^2x}{dt^2} = -kx \Leftrightarrow m \frac{d^2x}{dt^2} + kx = 0$$

where

$$k = m\omega^2$$

Applying the Newton's Second Law, we obtain the differential equation of a simple harmonic motion along  $\overrightarrow{OX}$  axis is then:

$$\frac{d^2x}{dt^2} + \frac{k}{m}x = 0 \quad \Leftrightarrow \quad \frac{d^2x}{dt^2} + \omega^2x = 0 \quad (120)$$

A solution of 120 is:

$$x = X_m \sin(\omega t + \varphi); \quad \text{but also} \quad x = X_m \cos(\omega t + \varphi) \quad (121)$$

## Definition and Description

The simplest kind of oscillation occurs when the restoring vector force  $\vec{F} = -k\vec{x} \Rightarrow F = -kx$  is proportional to the displacement vector  $\vec{x}$  from equilibrium position; the oscillatory motion is then called "***the Simple Harmonic Motion (SHM)***".

By definition, a particle moving along  $x$ -axis has a SHM when the particle's displacement  $x$  relative to the origin of the coordinate system is given by the relation

$$x = X_m \sin(\omega t + \varphi); \quad \text{or} \quad x = X_m \cos(\omega t + \varphi) \quad (122)$$

In this presentation we will use

$$x = X_m \sin(\omega t + \varphi)$$

## Definition and Description (Cont.)

The **instantaneous velocity**,  $v$ , (vibration speed) of a particle in a simple harmonic motion is by definition:

$$v = \frac{dx}{dt} = \omega X_m \cos(\omega t + \varphi) \quad (123)$$

The **instantaneous acceleration**,  $a$ , of a particle in a simple harmonic motion is given by:

$$a = \frac{d^2x}{dt^2} = -\omega^2 X_m \sin(\omega t + \varphi) = -\omega^2 A \sin(\omega t + \varphi) \quad \Leftrightarrow \quad a = -\omega^2 x \quad (124)$$

The relationship between instantaneous velocity  $v$ , instantaneous position  $x$  and the amplitude  $X_m$  can be given by:

$$\Rightarrow \quad v = \pm \omega \sqrt{X_m^2 - x^2} \quad (125)$$

## Energy of a Particle in SHM

The **kinetic energy** of a particle for the translation motion with a velocity  $v$  is given by:

$$K = \frac{1}{2}mv^2,$$

for the particle in simple harmonic motion, the position and the velocity are given by  $x = X_m \sin(\omega t + \varphi)$  and  $v = \frac{dx}{dt} = \omega X_m \cos(\omega t + \varphi)$  respectively.

$$\Rightarrow K = \frac{1}{2}m\omega^2 X_m^2 \cos^2(\omega t + \varphi),$$

but  $\forall \theta \cos^2 \theta = 1 - \sin^2 \theta$ . So, **the kinetic energy  $K$  of a particle** in a simple harmonic motion becomes:

$$K = \frac{1}{2}m\omega^2 X_m^2 [1 - \sin^2(\omega t + \varphi)] = \frac{1}{2}m\omega^2 (X_m^2 - x^2) \quad (126)$$

## Energy of a Particle in SHM (Cont.)

The **potential energy**  $U$  of a particle in a simple harmonic motion becomes:

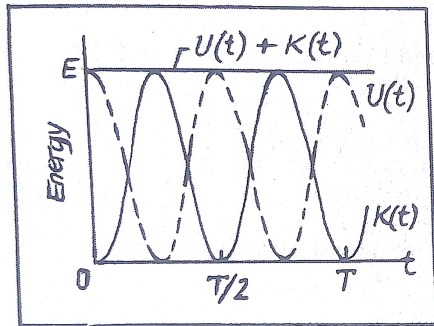
$$U = \int_0^U dU = \int_0^x kx dx \Rightarrow U = \frac{1}{2}kx^2 = \frac{1}{2}m\omega^2 x^2 \quad (127)$$

By definition, the **total mechanical energy**  $E$  of a particle equals to the sum of its kinetic energy and potential energy.

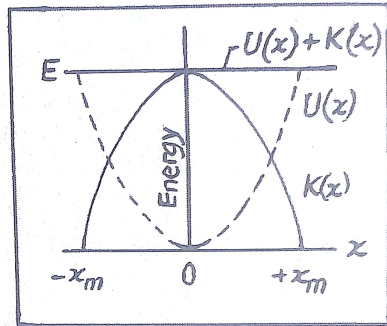
$$E = K + U = \frac{1}{2}m\omega^2 (X_m^2 - x^2) + \frac{1}{2}m\omega^2 x^2 = \frac{1}{2}m\omega^2 X_m^2 = \text{Constant} \quad (128)$$

## Energy of a Particle in SHM (Cont.)

Since  $m$ ,  $\omega$  and  $X_m$  are constant, the total mechanical energy  $E$  of a particle in a simple harmonic motion is also constant; we say that  $E$  **is conserved**.



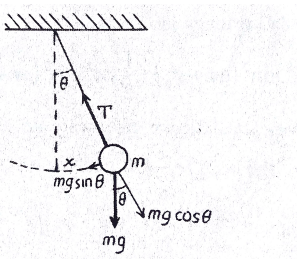
(a)



(b)

A point mass  $m$  (bob of the pendulum) hanging by a massless string of length  $l$  is an idealized system for a **simple pendulum**.

For the small angles  $\sin(\frac{x}{l}) \approx \frac{x}{l}$ ,  
Equation 129 becomes



$$F \approx \frac{mgx}{l} \quad (130)$$

$$m\omega^2 = \frac{mg}{l} \quad (131)$$

or

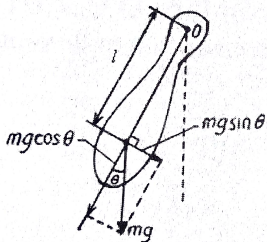
$$\omega = \sqrt{\frac{g}{l}} \quad (132)$$

$$F = -mg \sin\left(\frac{x}{l}\right) \quad (129)$$

From 130, we get

$$m \frac{d^2x}{dt^2} = -mg \left(\frac{x}{l}\right) \quad \text{or} \quad \frac{d^2x}{dt^2} + g \left(\frac{x}{l}\right) = 0 \quad (133)$$

A **physical pendulum** is formed by any irregular hanging object that oscillates about a fixed axis that does not pass through the center of the mass of the object.



$$\tau_{net} = I\alpha = F_{net}L = -mgL \sin \theta \quad (134)$$

$$\Rightarrow I \frac{d^2\theta}{dt^2} = -mgL \sin \theta \quad (135)$$

For the small angles, Equation (135) becomes the differential equation of the oscillating system:

$$\frac{d^2\theta}{dt^2} = -\frac{mgL\theta}{I} \quad (136)$$

The angular frequency  $\omega$  is given by

$$\omega = \sqrt{\frac{mgL}{I}} \quad (137)$$

The motion of a physical pendulum can be described by the relation between the **net external torque**  $\tau_{net}$  applied to the system and the **angular acceleration**  $\alpha$  of the system

# Introduction

The idealized simple harmonic motion is **frictionless**. There are **conservative forces**, the total mechanical energy is constant, and a system set into motion will continue to **oscillate forever without a decrease in amplitude**.

Real-world systems always have some **dissipative forces**, and **oscillations do die out with time**; unless we provide some means for replacing the dissipated mechanical energy.

In all those cases, the decrease in amplitude caused by dissipative forces is called "**damping**", and the corresponding motion is called "**damped oscillations**".

The amplitude  $X_m$  of oscillations decreases progressively in time, because the loss of energy due to the body's **motion in a viscous medium**, that opposes a **resistant force** to the velocity vector  $\vec{v}$ .

# Viscous Friction

⇒ When a body is moving in a viscous medium with a small velocity  $v$ , it is proved that **the force of the viscous friction**  $\vec{F}_v$  is quite proportional to the body's velocity  $v$ , but opposite to the motion's direction.

$$\vec{F}_v = -b\vec{v} \quad \Rightarrow \quad F_v = -bv = -b\frac{dx}{dt} \quad (138)$$

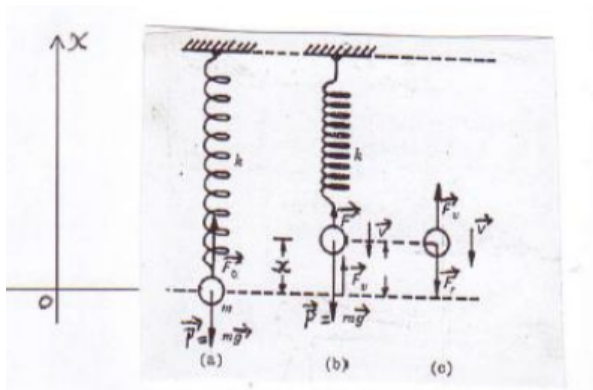
where  $b$  is a constant that depends on the body's **shape** and **size**, but it is also depends on the **viscous fluid's nature**.

⇒ When the flow is **turbulent** (formation of whirlpools behind the solid body), the force of viscous friction  $\vec{F}_v$  increases quickly in function of the body's velocity  $\vec{v}$  and its magnitude becomes:

$$F_v = b_1v + b_2v^2 + b_3v^3 + \dots \quad (139)$$

# Damped Mechanical Oscillation Equations

Consider an oscillator **mass-spring ( $m-k$ )**, composed by a solid sphere of mass  $m$  hanged on a spring of force constant  $k$  (the strength of the spring).



At the equilibrium  $\vec{F}_0 = -\vec{P} \Rightarrow F_0 = P = mg$

## Damped Mechanical Oscillation Equations (Cont.)

When the solid sphere is oscillating at a velocity  $v$  into the viscous fluid, of coefficient of viscous friction  $b$ , and if we give to the sphere an initial amplitude  $X_{m0}$ ; so, the sphere's oscillations will be necessary damped.

Applying the Newton's Second law for Dynamics of translation and considering that the Archimedes' force is negligible, we get differential equation of the damped oscillations.

$$\frac{d^2x}{dt^2} + \frac{b}{m} \frac{dx}{dt} + \frac{k}{m}x = 0 \quad (140)$$

or

$$\frac{d^2x}{dt^2} + 2\delta \frac{dx}{dt} + \omega_0^2 x = 0 \quad (141)$$

where  $\delta$  is **the damping factor** and  $\omega_0$  is **the angular frequency of a simple harmonic motion** associated to the damped oscillations.

## Damped Mechanical Oscillation Equations (Cont.)

A solution of equation (141) is

$$x = X_m(t) \cos(\omega t + \varphi) \quad (142)$$

The amplitude  $X_m(t)$  of damped oscillations decreases exponentially in function of time according to the following equation:

$$X_m = X_m(t) = X_{m0} e^{-Ct} \Rightarrow x = X_{m0} e^{-Ct} \cos(\omega t + \varphi) \quad (143)$$

$\Rightarrow C$  and  $\omega$  must be related to the constants of the ***m-k-b*** system, whereas  $X_{m0}$  and  $\varphi$  depend more on the initial abscissa  $x_0$  and velocity  $v_0$  ( $t = 0$ )

The ***angular frequency***  $\omega$  of damped oscillations is given by:

$$\omega = \sqrt{\omega_0^2 - \delta^2} = \omega_0 \sqrt{1 - \left(\frac{\delta}{\omega_0}\right)^2} \quad (144)$$

# Introduction

If we need that **the amplitude** of oscillations **remains constant**, we must continuously give to the system an energy that equals to the lost energy at each period, by applying a force that varies with time in a periodic way and with a definite frequency. In that case, oscillations are called **forced oscillations or driven oscillations**.

The differential equation of forced or driven oscillations is

$$\frac{d^2x}{dt^2} + 2\delta \frac{dx}{dt} + \omega_o^2 x = \omega_o^2 X_{me} \cos(\omega_e t) \quad (145)$$

Its solution is given by:

$$x = X_m \cos(\omega_e t + \theta) \quad (146)$$

Where  $X_m$  and  $\theta$ , are the amplitude and the phase angle (initial angular phase) of forced or driven oscillations, respectively.

➤ The **amplitude  $X_m$  of forced oscillations** is given by:

$$X_m(\omega_e) = \frac{X_{me}}{\sqrt{\left[1 - \left(\frac{\omega_e}{\omega_o}\right)^2\right]^2 + \left(\frac{2\delta}{\omega_o}\right)^2 \left(\frac{\omega_e}{\omega_o}\right)^2}} \quad (147)$$

➤ The **phase angle  $\theta$  of forced oscillations** is given by:

$$\theta = \arctan \frac{\frac{2\delta}{\omega_o} \left(\frac{\omega_e}{\omega_o}\right)}{\left[\left(\frac{\omega_e}{\omega_o}\right)^2 - 1\right]} \quad (148)$$

❶ The amplitude  $X_m$  of force oscillations takes a maximum value  $X_M$  when  $\omega_e = \omega_m$  and quite less than  $\omega_0$ ; so that for  $X_m = X_M$ , we get

$$\omega_m = \omega_0 \sqrt{1 - 2 \left( \frac{\delta}{\omega_0} \right)^2} \quad (149)$$

✎ There will be no resonance ( $\omega_m = 0$ ), when

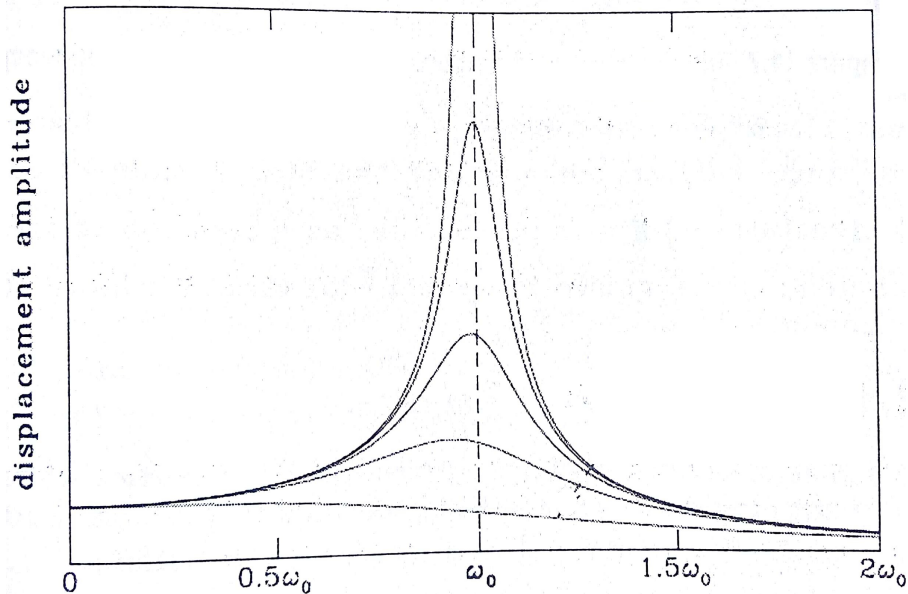
$$\left( \frac{\delta}{\omega_0} \right) \geq \frac{1}{\sqrt{2}}$$

✎ If the damping  $\delta$  is very small, the  $\left( \frac{\delta}{\omega_0} \right) \simeq 0$  and  $\omega_m \simeq \omega_0$  and we say that the system is in resonance

❷ We call **the fact of quality** of forced oscillator ( $m - k - b$ ), the quantity:

$$Q = \frac{\omega_0}{\omega_e} \simeq \frac{\omega_0}{2\delta} = \frac{\omega_0 m}{b} \quad (150)$$

✎ If  $b$  is small, then  $Q$  is high and the system will be more *resonant*; we say that **the resonance is sharp**. The resonance will be **flat** in the opposite case.



# Introduction

The mechanical waves is produced when the section of the propagation medium is displaced from its position of equilibrium. Then, cause of the elastic properties of that medium, that section starts to vibrate around the equilibrium position.

The disturbance (deformation) produced travels through the medium, from one section to the closer one, putting it in the same vibration but at successively later time. We say that there is **vibration propagation** or simply **a progressive wave**.

# Types of Waves

☞ According to direction of particles' oscillations and the wave propagation direction, mechanical waves can be **transverse** or **longitudinal**.

- A wave is **transverse** when on its passage, the particles of the propagation medium oscillate perpendicular to the wave direction of propagation.
- A wave is **longitudinal** when on its passage, the particles of the propagation medium oscillate in the same direction as the wave direction of propagation.

☞ According to the number of space dimensions in which the wave propagates, waves can be:

- **One-dimension waves:** waves along a stretched rope or spring
- **Two-dimension waves:** concentric circular waves (crest and valleys) on a liquid surface or on a drum membrane
- **Three-dimension waves:** sound waves, seismic waves in solids, light waves, electromagnetic waves used in Telecommunication.

# Propagation of waves

Waves propagation occurs by **short pulses** produced and sustained at a regular time rate; in that case, the waves are called **progressive waves** and can travel through the medium with a velocity  $\vec{v}$ , called the **wave celerity**;

When two wave pulses,  $y_1(x, t)$  and  $y_2(x, t)$  vibrating in the same direction and travelling in opposite direction overlap, the total displacement  $y(x, t)$  of any point at any time is obtained by adding the separate displacements, according to the principle of superposition  $y = y_1(x, t) + y_2(x, t)$ .

Note that after the overlapping, each wave pulse follows its motion without any deformation.

# Reflection and refraction of wave pulse

When a traveling wave pulse reaches a rigid obstacle:

- its part 1 is transmitted;
- its part 2 is reflected; and
- its part 3 (small) is absorbed.

At the boundary of two media with different inertia ( $N_1$  and  $N_2$ ), and incident pulse sets up:

- a weak refracted wave pulse, vibrating in the same direction as the incident wave pulse; if the second medium's inertia is the biggest:  
 $N_1 < N_2 \Rightarrow v_1 > v_2$
- a weak refracted wave pulse and a strong reflected wave pulse, both vibrating in the same direction as the incident wave pulse; if the second medium's inertia is smallest:  $N_1 > N_2 \Rightarrow v_1 < v_2$

## wave speed

Consider a rope that is characterized by its mass per unit length, stretched by the force  $\vec{T}$  and a transverse wave along that rope, whose celerity is  $\vec{v}$

During the time interval  $\Delta t$ , the wave front travels an elementary length  $\Delta l$  along  $\vec{\Delta x}$ -direction and this is done through the stretched rope;

The motion of the elementary mass  $\Delta m$  of the rope that is reached by the wave front, and which is extended along  $\Delta l$ , is composed by two perpendicular motions (vibration and propagation):

Uniformly varying (along  $\Delta y$ -direction of vibration)

$$a_y = \frac{T \sin \theta}{\Delta m} \Rightarrow v_y = a_y \Delta t = \frac{T \sin \theta}{\Delta m} \Delta t \quad (151)$$

## wave speed (cont.)

Uniform motion (along  $\Delta x$ -direction of propagation)

$$\Delta l = v_x \Delta t \Leftrightarrow \Delta t = \frac{\Delta l}{v_x} \quad (152)$$

Putting 152 in 151, where  $v_x = v$ , the wave celerity along the string, we get:

$$v_y = \frac{T \sin \theta}{\Delta m} \times \frac{\Delta l}{v} \quad \text{with} \quad v_y = v \tan \theta \Rightarrow v^2 = v_x^2 = T \times \frac{\Delta l}{v} \times \cos \theta$$

The wave celerity  $v$  along a stretched string or rope is given by the following general expression:

$$v = \sqrt{\frac{T}{\mu} \cos \theta} \quad (153)$$

## Quick Quiz

A transverse traveling wave on a taut wire has an amplitude of 0.200 mm and a frequency of 500 Hz. It travels with speed of 196 m/s.

- (a) Write an equation in SI units of the form  $y = A \sin(kx - \omega t)$  for this wave.
- (b) The mass per unit length of this wire is 4.10 g/m. Find the tension in the wire.

# Harmonic Waves (HW)

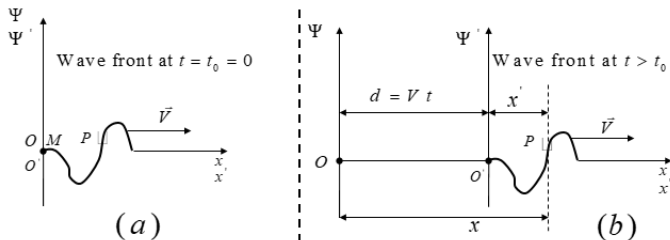
a mechanical wave is a **harmonic** or **sinusoidal wave** when it comes from the propagation of a SHM in a material medium.

Every harmonic wave is characterized by the following physical quantities:

- Its **amplitude**: the maximum value that can take the physical quantity, associated to the wave;
- Its **wavelength** or **spatial period**,  $\lambda$ , that represents the distance covered by the wave during a period of time, and is expressed in meters;
- Its **period**,  $T$  or **the time duration of one complete vibration**, expressed in seconds; and
- Its **frequency**,  $f$  which indicates the number of vibrations or cycles per second; it is expressed in hertz (Hz).

# Harmonic function

Consider two photos of a HW, taken at two different times on the same rope as illustrated in the figures (a) and (b):



At time  $t = t_0 = 0$ , the physical quantity associated to the wave propagation is described by a wave function:  $\Psi(x, t_0) = \Psi(x, 0) = \Psi(x)$  with respect to the frame of reference whose origin is  $O$ ;

## Harmonic function (Cont.)

Another frame of reference whose origin is  $O'$ , having the same motion as the wave, but with  $O \equiv O'$  at  $t = t_0 = 0$  (examples of deformation for a vibrating rope, or compression of a spring).

At any time  $t$ :  $\Psi(x, t) = \Psi(x') = \Psi(x - vt) = \Psi_m \sin[k(x - vt)]$ .

As the function **sine** is periodic, we have

$$\Psi(x, t) = \Psi(x + \lambda, t) = \Psi(x, t + T)$$

Therefore, the wave function for a HW is expressed:

$$\begin{aligned} \Psi(x, t) &= \Psi_m \sin[k(x + \lambda) - kvt] = \Psi_m \sin[k(x - vt) + 2\pi] \\ &= \Psi_m \sin[kx - kv(t + T)] = \Psi_m \sin(kx - \omega t) \end{aligned} \quad (154)$$

## Harmonic function (Cont.)

Where the constant  $k$  stands for **wave number** with relationship

$$\lambda k = 2\pi \Leftrightarrow k = \frac{2\pi}{\lambda} \quad (155)$$

The angular frequency  $\omega$  of harmonic waves and expressed in radians per second (1 rad/s) becomes:

$$\omega = \frac{2\pi}{\lambda} v = kv = \frac{2\pi}{T} = 2\pi f \quad (156)$$

The relationship between the wavelength  $\lambda$ , the period  $T$  and the frequency  $f$  of a harmonic wave is given by:

$$\lambda = vT = \frac{v}{f} \quad (157)$$

# Energy Transport by Waves

In a wave propagation, the energy given to the wave source at the beginning propagates through the material medium at a constant speed  $v$ .

Compared to the expression of the total mechanical energy of a SHM; the total mechanical energy  $E$ , carried on by a harmonic wave is also:

$$E = \frac{1}{2} m \omega^2 \psi_m^2 = \frac{1}{2} m \omega^2 A^2 \quad (158)$$

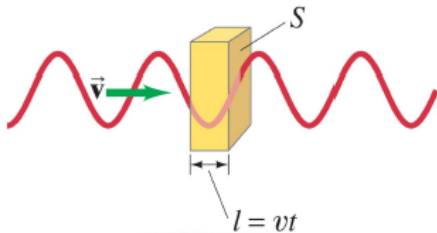
The **density of energy**  $\varepsilon$  carried on by a harmonic wave equals to the total energy per unit volume  $V_o$ :

$$\varepsilon = \frac{E}{V_o} = \frac{1}{2} \frac{m}{V_o} \omega^2 \psi_m^2 = \frac{1}{2} \rho \omega^2 \psi_m^2 = \frac{1}{2} \rho \omega^2 A^2 \quad (159)$$

## Energy Transport by Waves (Cont.)

The power  $P$  (in  $W$ ) carried on by the harmonic wave becomes:

$$P = \frac{E}{t} = \frac{\varepsilon V_o}{t} = \varepsilon v S = \frac{1}{2} \rho \omega^2 v S \Psi_m^2 = \frac{1}{2} (\rho \omega^2 v S) A^2 \quad (160)$$



The **wave intensity**  $I$  is the energy carried on by a wave per unit time and per unit cross-section area.

$$I = \frac{P}{S} = \varepsilon v = \left( \frac{1}{2} \rho \omega^2 v \right) \Psi_m^2 = \left( \frac{1}{2} \rho \omega^2 v \right) A^2 \quad (161)$$

# Linear Wave Equation

The propagation of an harmonic wave through a medium, along an axis  $\overrightarrow{OX}$  at a celerity  $\vec{v}$ , is described by the following general wave function:

$$\Psi(x, t) = \Psi(x \mp vt) \quad (162)$$

However, it can also be described by a differential wave equation for a wave that propagates along the  $\overrightarrow{OX}$  axis.

Suppose that  $u = x \mp vt$ , we get

$$\Psi(x, t) = \Psi(u) = \Psi(x \mp vt) \quad (163)$$

# Linear Wave Equation (Cont.)

By derivation of Equation 163, successively with respect to  $x$  then to time  $t$ , we have:

$$\frac{\partial \Psi}{\partial x} = \frac{\partial \Psi}{\partial u} \frac{\partial u}{\partial x} = \frac{\partial \Psi}{\partial u} \quad \text{and} \quad \frac{\partial^2 \Psi}{\partial x^2} = \frac{\partial^2 \Psi}{\partial u^2} \quad (164)$$

then

$$\frac{\partial \Psi}{\partial t} = \frac{\partial \Psi}{\partial u} \frac{\partial u}{\partial t} = \mp v \frac{\partial \Psi}{\partial u} \quad \text{and} \quad \frac{\partial^2 \Psi}{\partial t^2} = (\mp v)^2 \frac{\partial^2 \Psi}{\partial u^2} \frac{\partial u}{\partial t} = v^2 \frac{\partial^2 \Psi}{\partial u^2} \quad (165)$$

that is

$$\frac{\partial^2 \Psi}{\partial u^2} = \frac{1}{v^2} \frac{\partial^2 \Psi}{\partial t^2} \quad (166)$$

# Linear Wave Equation (Cont.)

Comparing 165 and 166, we get:

$$\frac{\partial^2 \psi}{\partial x^2} = \frac{1}{v^2} \frac{\partial^2 \psi}{\partial t^2} \quad (167)$$

The equation (2.52) is ***the one-dimension wave differential equation*** of a wave that travels along the  $\overrightarrow{OX}$  axis.

In general, the three-dimension wave differential equation is:

$$\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} + \frac{\partial^2 \psi}{\partial z^2} = \frac{1}{v^2} \frac{\partial^2 \psi}{\partial t^2} \quad (168)$$

# Outline

## 9 OPTICS AND WAVES

- Introduction
- Properties of Light wave
- Polarization of light
- The lens, Mirror and Image Formation

Solar rays on the earth, Stars in the sky, you can see your face in morning, all these are optical phenomena. In this section, we are going to study Wave Optics and Geometric Optics. We will start with the waves aspect of light, light propagation in matter, and we close this study with the information carried by the light ray.

Having successfully completed this section. you should be able to:

-  Discuss the wave character of light;
-  Identify the changes when light enters a medium;
-  Explain with ray diagrams the formation of an image using plane and spherical mirrors;
-  Illustrate the formation of images using the technique of ray tracking.

# Definition

If you have ever looked at the reds, blues, and greens in a sunlit soap bubble and wondered how straw-colored soapy water could produce them, you have hit upon one of the many phenomena that can only be explained by the wave character of light. The same is true for the colors seen in an oil slick or in the light reflected from a compact disc. These and other interesting phenomena, such as the dispersion of white light into a rainbow of colors when passed through a narrow slit, cannot be explained fully by geometric optics. In these cases, light interacts with small objects and exhibits its wave characteristics. The branch of optics that considers the behavior of light when it exhibits wave characteristics is called wave optics. It is the topic of this chapter.



# Nature of light waves

We know that visible light is the type of electromagnetic wave to which our eyes respond. Like all other electromagnetic waves, it obeys the equation

$$c = f\lambda \quad (169)$$

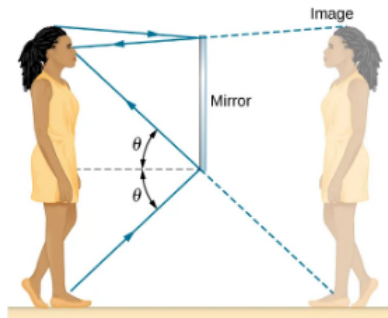
where  $c=3 \times 10^8 \text{ m/s}$  is the speed of light in vacuum,  $f$  is the frequency of the electromagnetic waves, and  $\lambda$  is its wavelength.

Light has wave characteristics in various media as well as in a vacuum.

When light goes from a vacuum to some medium, like water, its speed and wavelength change, but its frequency  $f$  remains the same. (We can think of light as a forced oscillation that must have the frequency of the original source.) The speed of light in a medium is  $v = \frac{c}{n}$ , where  $n$  is its index of refraction. If we divide both sides of equation  $c = f\lambda$  by  $n$ , we get  $\frac{c}{n} = v = \frac{f\lambda}{n}$ . This implies that  $v = f\lambda_n$ , where  $\lambda_n$  is the wavelength in a medium and that with  $\lambda_n = \frac{\lambda}{n}$

# Reflection of light wave

When we see ourselves in a mirror, it appears that our image is actually behind the mirror. We see the light coming from a direction determined by the law of reflection. The angles are such that our image is exactly the same distance behind the mirror as we stand away from the mirror. If the mirror is on the wall of a room, the images in it are all behind the mirror, which can make the room seem bigger.



# laws of reflection(Examples)

## Example No. 15

- ① Show that when light reflects from two mirrors that meet each other at a right angle, the outgoing ray is parallel to the incoming ray
- ② A flat mirror is neither converging nor diverging. To prove this, consider two rays originating from the same point and diverging at an angle  $\theta^0$ . Show that after striking a plane mirror, the angle between their directions remains  $\theta^0$

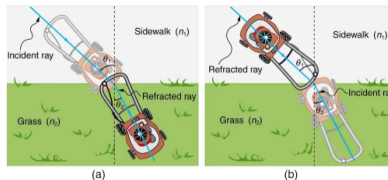
# laws of reflection(Examples)

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# Refraction of light wave

Figure below shows how a ray of light changes direction when it passes from one medium to another. As before, the angles are measured relative to a perpendicular line to the surface at the point where the light ray crosses it. (Some of the incident light will be reflected from the surface, but for now we will concentrate on the light that is transmitted.) The change in direction of the light ray depends on how the speed of light changes. The change in the speed of light is related to the indices of refraction of the media involved. In the situations shown in Figure, medium 2 has a greater index of refraction than medium 1. This means that the speed of light is less in medium 2 than in medium 1.



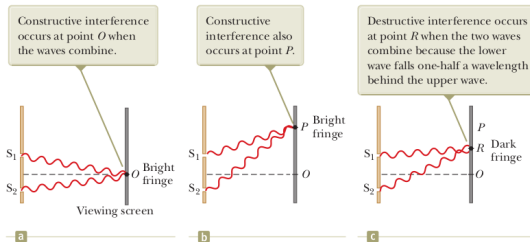
## laws of refraction(Examples)

### Example No. 16

- 1 Find the index of refraction for medium 2 in Figure above (a), assuming medium 1 is air and given the incident angle is  $30^0$  and the angle of refraction is  $22^0$ .
- 2 Suppose that in a situation like that in Example No. 16 (1) light goes from air to diamond and that the incident angle is  $30^0$  . Calculate the angle of refraction  $\theta$  in the diamond,  $n$  for diamond is 2.419.

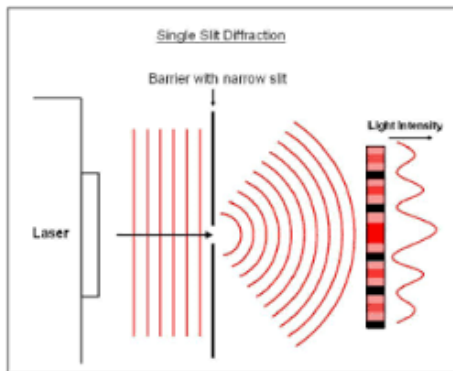
# Interference of light

In this subsection, we studied the waves in interference model and found that the superposition of two mechanical waves can be constructive or destructive. In constructive interference, the amplitude of the resultant wave is greater than that of either individual wave, whereas in destructive interference, the resultant amplitude is less than that of the larger wave. Light waves also interfere with one another. Fundamentally, all interference associated with light waves arises when the electro- magnetic fields that constitute the individual waves combine.



# Diffraction of light

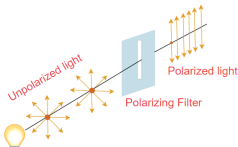
When plane light waves pass through a small aperture in an opaque barrier, the aperture acts as if it were a point source of light, with waves entering the shadow region behind the barrier. In this subsection, we investigate the features of the diffraction pattern that occurs when the light from the aperture is allowed to fall upon a screen.



# Polarization of light

If you were to leave your house on a hot, sunny day, you would undoubtedly wear sunglasses. This is because the light emitted by the sun is unpolarized light and the sunglasses we wear transform the unpolarized light. Polarized light is light in which the electric field vector of the light is in the same phase and is perpendicular to the propagation of the light wave. The process of converting unpolarized light into polarized light is called polarization.

When the electric field vectors are restricted to a single plane by filtration, then the light is said to be polarized with respect to the direction of propagation and all waves vibrate in the same plane.



# Image Formation

## Example No. 17

Discuss the following topics.

Images Formed by  
Flat Mirrors

Images Formed by Spherical  
Mirrors

Images Formed by  
Refraction

Images Formed by  
Thin Lenses

Lens Aberrations

The Camera

The Eye

The Simple Magnifier

The Compound Microscope

The Telescope

# The End